RESEARCH REPORT

Allocating AI Compute for Global Educational Access

Evidence-bound methods for language, stage, subject, and accessibility allocation

A global portfolio model for language, shared-core, accessibility, and open-curriculum decisions

Evidence freeze - 2026-09-04

Abstract

The evidence freeze represents 1,077 authority targets: 204 source-authorized direct-person targets, 2 separately comparable stage-opportunity targets, and 871 visible unranked or contextual targets. A common-prior reference order places Hindi and Bahasa Indonesia first among comparable person-denominator rows; Simplified Standard Written Chinese leads the separate stage-opportunity lane, and Japanese remains a targeted tertiary, research, and accessibility opportunity. The full Top 100, high-consequence evidence gaps, signed-language and accessibility portfolio, displacement portfolios, and interlanguage experiments are all published rather than reduced to one league table. Sensitivity analysis shows that the direct order is highly population-dominant because most exact scarcity and academic-overlap factors remain unmeasured; its correct use is transparent high-reach prioritization plus target-specific package evidence, not a claim of observed global need.

1. Introduction

The relevant global shortage is not simply a shortage of text. Educational access depends on whether a learner can obtain, understand, navigate, retain, reuse, and study appropriate material at the required stage and subject. A book can exist while remaining unusable because it is written in an unfamiliar academic language, addresses a different curriculum, assumes unavailable prerequisites, omits exercises or solutions, requires continuous connectivity, cannot be read by assistive technology, or is legally unavailable for adaptation and redistribution. Conversely, a language with extensive educational publishing at one stage can still have major gaps in vocational training, postgraduate methods, advanced research, accessible formats, or independent self-study.

UNESCO's current multilingual-education guidance repeats the broad estimate that roughly 40% of people lack access to education in a language they speak and understand fluently (UNESCO, 2025). That figure is a global context, not a current language-by-language census. Its historical lineage includes Walter and Benson's (2012) tabulation of 97 languages with more than ten million speakers: a later published reproduction reports that 45 were not used in education and represented more than one billion speakers, while 52 were used in education (Spolsky, 2021). The publication does not expose a verified named appendix of all 97 languages. Those aggregates must therefore not be reverse-engineered into a modern target list. They are also unrelated to Crawford et al.'s (2025) analytical sample of 96 alphabetic languages in Early Grade Reading Assessment data.

The World Bank's more recent language-of-instruction model addresses a different denominator. It estimated that approximately 37% of students in low- and middle-income countries - about 370 million of roughly one billion students in its modeled universe - were taught in a language other than the language they best spoke and understood, and it examined a scenario involving roughly 550 additional languages (World Bank, 2021). This is direct evidence that language alignment is a large educational variable. It is not evidence that all affected students need the same package, that translation alone resolves instruction, or that one national average can be assigned to every language community within a country.

AI changes what can be produced with a fixed budget. A sufficiently capable mathematical or scientific model can jointly translate and inspect meaning, terminology, formula bindings, assumptions, and argument structure. Independent model passes and deterministic source, formula, structure, link, and render checks can reduce error without making publication depend on an unavailable human reviewer. Yet production feasibility and educational value remain empirical questions. Model quality varies by exact direction, script, domain, corpus support, and tokenizer; a language with sparse public data may require wider uncertainty and more audit compute, but it does not thereby become less important.

This paper asks:

> Which exact language, script, signed-language, accessibility, and evidence-supported bridge interventions should receive standardized new AI compute to create the greatest increase in functional educational access?

The study answers that question in two primary orders. The first measures baseline unmet need without reference to production history or cost. The second estimates expected access gain per standardized from-scratch compute for a fixed open educational package. This separation prevents convenience, sunk work, local holdings, or prior project choices from being mistaken for properties of a population.

2. Outcomes and units of analysis

2.1 Functional access

Functional access is endpoint-specific. A learner has functional access when the material can be used for its stated educational purpose under the declared delivery conditions. The study distinguishes at least:

1. autonomous or self-study use;
2. instruction-mediated use;
3. vocational or professional task use; and
4. research or reference use.

These endpoints are not interchangeable. A free-to-read research article may provide reference access without providing a sequenced course. A school textbook may support classroom instruction while being incomplete for independent study. An open license creates adaptation and redistribution capacity but does not prove that a file is usable offline, accessible, complete, or pedagogically effective. A closed but lawful and no-cost national textbook may reduce present functional scarcity on its exact route while contributing no open-adaptation capacity.

2.2 Exact language targets

The language target is defined by variety or register, script and orthography, territory or community, and educational function. Spoken Putonghua is not a population estimate for readers of simplified Standard Written Chinese. Modern Standard Arabic is a learned written and educational register rather than an ordinary first-language denominator. Filipino educational exposure is not Tagalog first-language use. Punjabi census counts do not identify Gurmukhi and Shahmukhi literacy. Named sign languages are independent languages; a national hearing-loss count is not a sign-language-user count.

Macro labels are retained as contextual ceilings or package hypotheses when useful, but their populations are not copied into child targets. Transnational languages are assembled only from country-disjoint cells with compatible definitions. When a shared core may reduce production cost across standards - such as parallel Bangla editions in Bangladesh and India, Gurmukhi and Shahmukhi Punjabi, Persian-Dari-Tajik standards, or national Kiswahili profiles - the package relationship is represented separately from exact-language need.

2.3 Educational stages and subjects

Education is represented from ISCED 0 through ISCED 8 and adult or professional learning (UNESCO Institute for Statistics, 2012). The stage map includes:

- early childhood and caregiver-facing learning;
- primary literacy, numeracy, science, and bridge-language learning;
- lower- and upper-secondary mathematics, sciences, computing, writing, health, and civic or practical subjects;
- technical and vocational education;
- undergraduate gateway and degree sequences;
- postgraduate research methods and specialist study;
- advanced research texts, monographs, terminology infrastructure, and scholarly communication; and
- adult literacy, health, finance, management, agriculture, climate, digital skills, and workforce learning.

Foundational learning is not treated as the only form of education. Advanced mathematics, research methods, scholarly reading, and specialist reference materials serve smaller cohorts but can have substantial research and institutional effects. They are estimated at their actual opportunity populations rather than inflated to whole-language totals or erased by the presence of basic school content.

3. Evidence universe

3.1 Candidate generation

Candidate inclusion does not depend on population size, national income, official status, model support, prior project selection, or evidence completeness. The census sweeps all regions and includes large official languages, minoritized and Indigenous languages, refugee and diaspora languages, signed languages, and accessibility or delivery overlays. A candidate may remain unranked because evidence is insufficient; it does not disappear and is not assigned zero need.

The candidate record preserves exact target tags, scripts, orthographies, communities, stages, modalities, source identities, population classes, and all incompatibilities. Population classes include direct first-language counts, home use, reported ability, stage enrolment, direct support-need cohorts, territorial exposure ceilings, displacement cohorts, disability proxies, households, person-times, and learner-years. Only compatible person measures enter a common person order. Households, percentages, FTEs, and person-times are never silently converted to persons.

3.2 Population and denominator discipline

All arithmetic population values are normalized to persons while retaining the original published unit and precision. A low/base/high triplet is an uncertainty interval only when all three values describe the same construct, universe, and period. Different constructs cannot be relabeled as interval endpoints. For example, Japanese nationals and all residents are distinct exposure definitions, not the low and high values of a Japanese-reader population. A rounded speaker estimate and a national resident total are not a valid Turkish-language interval.

Each additive cell has an immutable population source, denominator, person basis, partition identity, and exclusivity statement. Renaming the cell or changing the point estimate cannot turn one population into two additive populations. A total-population cell cannot be added to its age, stage, disability, migration, or enrolment subsets. Alternative estimates form uncertainty scenarios rather than extra beneficiaries.

3.3 Learning, completion, and language alignment

Learning poverty, below-minimum proficiency, noncompletion, and first/home-language alignment describe different events. Country values are retained as national context unless a source or explicit transport model connects them to the exact target community and learner stage. Applying the same country average to every language can conceal within-country exclusion, particularly for Indigenous, minoritized, displaced, rural, or signed-language learners. Where only an ecological proxy is available, the point factor remains missing and the analysis carries a broad partial-identification interval.

The evidence model records the numerator event, denominator stratum, conditioning event, endpoint, and horizon for each factor. Marginal deficits are never labeled as a conditional chain merely because they appear in a convenient order. Where joint distributions are unavailable, overlapping need dimensions use conservative Fréchet union and intersection bounds.

3.4 Academic-language non-overlap

Conversational knowledge of English, French, Spanish, Portuguese, Russian, Arabic, Mandarin, or Hindi is not equivalent to independent academic learning through that language. School foreign-language thresholds, census declarations, commercial indices, bibliometric publication language, and discipline-specific reading are different measures. The evidence register therefore keeps academic-language access missing unless the source matches the target population, stage, task, and denominator.

This rule matters for large national languages as much as for minoritized ones. Available evidence identifies substantial school-stage English thresholds in Japan and Korea, English assessment burdens in the Philippines, foundational English gaps in parts of South Asia, and large reading-time burdens for researchers working outside their strongest language (Amano et al., 2023). None supplies one global academic-English percentage. Missing academic-language data increases uncertainty; it does not reduce need.

3.5 Existing resources and openness

The resource audit separates:

- localized interface or catalogue presence;
- usable components;
- coherent courses or sequences;
- stage-wide or cross-stage functional routes;
- no-cost access;
- unchanged redistribution;

- adaptation and redistribution rights;
- downloadable or offline operation;
- accessibility conformance; and
- independent-study completeness, including exercises, feedback, and solutions.

International catalogues are queried under identical rules, while official and country-native platforms are inspected to prevent an Anglophone catalogue from being treated as a census of local supply. Repository counts are evidence of discoverability, not curriculum coverage. Strong national supply is a negative control only for the exact stage, subject, language, and access property it demonstrates. It cannot erase open-license, offline, accessibility, advanced, vocational, adult, or research gaps.

Open availability is mandatory for newly commissioned packages. UNESCO's OER Recommendation defines the legal capability sought: no-cost access together with reuse, repurposing, adaptation, and redistribution (UNESCO, 2019). Technical usability is audited separately.

3.6 Connectivity and accessibility

Connectivity, device ownership, electricity, bandwidth, digital skills, and storage are distinct constraints. ITU estimated roughly 2.2 billion people offline in 2025, while the SDG 7 monitoring consortium estimated 666 million people without basic electricity access in 2023 (International Telecommunication Union, 2025; International Energy Agency et al., 2025). These groups overlap and cannot be added. Coverage also does not imply a personal device, affordable data, reliable charging, or the ability to store and navigate a large corpus.

Delivery is therefore modeled as separate branches: semantic static HTML with MathML, accessible EPUB, tagged and print-ready PDF, low-data audio with transcripts, plain-language editions, downloadable chapter bundles, local-server or removable-media packages, and named sign-language video where applicable. Disability prevalence is not copied into a format-specific beneficiary count. The relevant denominator is the exact functional barrier and intervention, and overlapping accessibility needs remain nonadditive.

4. Estimation method

4.1 Common baseline unmet-need estimand

For an exact language target l, let h index mutually exclusive cells defined by territory, complete strong-language and script-literacy profile, learner stage, subject/canon atom, and baseline access endpoint. Let D_h be the public person denominator and let $Y_h = 1$ mean that the person fails the fixed functional educational-access threshold. The needs-only quantity is

$$U_l = \sum_h a_{hl} D_h \theta_h ,$$

where $\theta_h = P(Y_h = 1)$ and a_{hl} assigns each person-stage atom to one exact language target in each bound or uncertainty draw. Assignment shares sum to one; overlapping language totals are not independent beneficiaries. A directly measured stage cohort is a valid denominator, so absence of a whole-language speaker census does not erase Japanese academic-language learners, written-Chinese stage cohorts, signed-language users, or any other directly defined population.

No package proposal, format branch, expected intervention effect, compute, local inventory, prior completion, or production history enters U_l. This is an aggregate access-burden estimand, not a moral ranking and not automatically a count of unique lifetime persons. Rights, vitality, cultural survival, prestige-domain value, and minimum-service principles are reported as separate decision lenses rather than hidden weights.

The study reports three versions of this same outcome. They differ in what the evidence identifies, not in whose need is valued.

4.2 L-C: direct conditional-chain identification

The strict lane accepts a product such as

$$D_h P(E_1 / h) P(E_2 / E_1 , h) \cdots P(Y_h / E_{k-1} , \ldots , E_1 , h)$$

only when every source supplies the required nested numerator, denominator, event identity, and immediate parent. This lane is useful for direct cohorts and genuine sequential filters. It rejects unrelated marginals and keeps the original executable safeguards.

The current evidence contains no globally complete L-C row. Learning shortfalls, nonattendance, language-of-instruction mismatch, lack of a usable resource, academic-language non-overlap, and delivery exclusion are usually overlapping mechanisms rather than a naturally nested sequence. Removing the conditionality checks merely to obtain 100 numbers would create fabricated precision. L-C therefore remains a high-evidence result and a validation benchmark, not the sole global ordering engine.

4.3 L-ID: assumption-light partial identification

Direct endpoint counts enter as direct burden observations and are not multiplied by missing component factors. Compatible component marginals yield Fréchet-Hoeffding union or intersection bounds; ecological or broader-cohort evidence does not enter an exact target without a declared transport relationship. For every target, cell bounds aggregate to $[U_l^-, U_l^+]$. Conservative rank bounds are

$$best_l = 1 + \#\{j : U_j^- > U_l^+\},$$

$$worst_l = n - \#\{j : U_j^+ < U_l^-\}.$$

The paper reports definite, possible, and outside Top-10/Top-100 membership under these bounds. Wide ties and overlapping rank sets are substantive findings about the evidence. They are not resolved through target names, region, evidence volume, or convenience.

4.4 L-M: common hierarchical model-conditional order

An ordered decision slate requires assumptions beyond the available direct cross-tabs. The reference model therefore treats θ_h as a latent parameter and uses the same stage-by-endpoint likelihood families, hyperpriors, measurement models, transport rules, and missingness model for every exact target. Direct assessments, construct-aligned counts, curriculum measures, language-of-instruction evidence, resource-route coverage, academic-language assessments, and delivery measures enter through source-specific observation models with their sampling and measurement error. Shared surveys and denominators retain correlated uncertainty.

In the current executable v3 snapshot, many country learning and connectivity fields are transported as an explicitly labelled ecological proxy: the model averages the available country indicators for the target's country and does not claim a target-language or target-stage cross-tab. The complete rule is recorded in `structured/FACTOR_TRANSPORT_REGISTRY_v1.json`. This transport supplies a model-conditional planning value only; the assumption-light L-ID result leaves the corresponding exact factor unidentified.

Target label, region, country-income class, prestige, official status, political status, evidence quantity, catalogue visibility, and project status are not predictors. Evidence tiers describe provenance and prior dependence; they do not change the prior mean or subtract from need. In the assumption-light L-ID lane, a blank observation remains unidentified and is not assigned zero or a midpoint. In the explicitly labelled L-M model-conditional lane, missing endpoint cells retain the preregistered common latent prior distribution; factor-level default values (for example, 0.5 for unresolved OER and accessibility, 0.4 for schooling-language and academic non-overlap, and 0.35 where no learning-context fields exist) are pinned in `structured/FACTOR_TRANSPORT_REGISTRY_v1.json` and are modelling assumptions, not observed values. A territorial ceiling constrains a latent membership share but is not relabelled as a reader count.

All target burdens and ranks are calculated jointly in each draw. The model-conditional Top K selects the K largest posterior expected burdens; exact ties crossing the boundary produce the complete tie set rather than an arbitrary target-ID order. For each of the 204 rankable person targets, the L-M output reports posterior mean, median, P05/P95 burden, L-ID bounds/status, model rank, Top-10 and Top-100 inclusion probabilities, evidence tier, and model status. The 2 exact stage-opportunity targets report model-conditional burden fields in their separate lane but no person-order rank or Top-K probability; the 871 unranked/context targets retain identity, source/authorization, and explicit status while rank/posterior fields remain blank. The persisted v3 sensitivity outputs report rank changes across four paired common-prior families and seven factor-removal/reference scenarios. This output is labelled a **model-conditional expected-need decision order**, not a table of directly measured global facts (Manski, 2003; Imbens & Manski, 2004; Gelman et al., 2013; Little & Rubin, 2019).

4.5 Package-specific access gain

For candidate package c, standardized horizon H, and stratum g:

$$G(c) = \sum_g P(g) E\left[N(i)\{A_c(i,H) - A_0(i,H)\}/g\right].$$

$N(i)$ is a prespecified binary need event or bounded severity measure. A_0 and A_c are baseline and supplied-package functional access under the same endpoint. Access gain may differ across worked examples, definition-heavy prose, proof prose, word problems, laboratory instructions, or research reference use. A language's need rank cannot be changed by proposing more packages.

4.6 Standardized new-compute efficiency

For a fixed package p and delivery branch f:

$$E(c,p,f) = \frac{G(c,p,f)}{C_{\text{new}}(c,p,f)}.$$

C_{new} is a standardized from-scratch fresh-equivalent compute estimate. It may depend on public, prospective production features: source size, target expansion under a pinned tokenizer, script and segmentation behavior, terminology density, formula and table density, diagrams, modality, accessibility branches, independent model-audit passes, deterministic QA, and rendering. Cached gross token counters are not converted directly into money, and prior completion or reusable local infrastructure is not an input.

The executable v3 comparison cannot identify G because no common causal effect of supplying the fixed package has been observed across the target populations. Its published `access_gain_per_compute_median` field is therefore a legacy field name for the explicitly model-conditional proxy U_l / C_{new}: unmet-need equivalent per standardized compute. It can compare planning envelopes while package effects remain unknown, but it must not be interpreted as observed learners helped, causal educational gain, or money saved. A later package-specific evaluation can replace U_l with a source-supported $G(c, p, f)$ without changing the needs-only order.

Compute evidence is classified on two axes. A **profile-input class** states whether the target representation itself has a directly aligned tokenizer measurement, a proved partial-identification set, a model-conditional prediction, no usable profile, or a mode for which written tokenization is inapplicable. A separate **total-cost class** states whether the entire standardized workflow cost was directly observed, partially identified by valid component bounds, or generated under a declared planning or predictive model. Direct measurement of a token-expansion ratio does not make an end-to-end translation, checking, accessibility, rendering, and packaging workload directly measured.

The current benchmark contains 203 non-English FLORES language-script profiles measured against one pinned corpus and tokenizer protocol. It contains zero directly executed end-to-end standardized package costs and zero total-cost rows with valid mathematical lower and upper bounds. Its 3,654 profile/canon/package totals combine a direct representation anchor with declared process scenarios and are therefore **measurement-anchored, model-conditional planning estimates**. The 18 cross-language p10/p50/p90 envelope cases are sensitivity controls, not target estimates or bounds. They receive no target key and cannot enter a point order.

Every final target receives exactly one profile-binding status. Exact variety, script or orthography, mode, and representation scope must match a pinned measurement row; a neighbouring language, macrolanguage, different script, or generic envelope cannot be substituted. Targets without an exact direct profile remain explicitly unavailable in the direct-profile subset. A complete model-conditional compute order may include them only through one frozen predictive protocol applied to the entire roster, with target labels, population, need, region, income, prestige, evidence quantity, local work, and project state excluded from the cost predictors. Predictive intervals remain predictive rather than being relabelled as bounds.

The compute order compares only interventions with a common benefit unit and complete common uncertainty draws. Partial favorable draws cannot create point ranks. Accessibility, audio, and signed-video branches have distinct production factors rather than being treated as free by-products of plain text.

4.7 Missing data, sensitivity, and value of information

Each empirical value has low, base, and high fields, source, denominator, reference year, geography, measure class, and confidence. Missing base values are not imputed with zero or a midpoint. In L-ID, interval-only candidates receive rank bounds but no point rank; in L-M, the same interval is sampled under the stated triangular model to produce a model-conditional point order while the raw interval remains visible. Candidates with no defensible interval remain visible in a data-priority lane.

The persisted v3 sensitivity outputs contain four paired common-prior families (816 rows) and seven factor-removal/reference scenarios (1,428 rows) over the 204 person targets. The release also publishes a target-level min/max-rank summary and scenario rank correlations. These analyses do not resolve missing target-specific factors; they show which conclusions are stable under the declared common-prior perturbations. A candidate is described as robust or definitely within a Top-K only when those claims follow from the supplied universe and uncertainty model.

The specification defines a value-of-information (VOI) evidence-acquisition schema/skeleton; its synthetic fixture is explicitly `SKELETON_NOT_COMPUTED_REQUIRES_PROSPECTIVE_DESIGN`, and no production VOI table is present in this v3 snapshot. This report therefore makes no claim that missing measurements have been ranked by expected regret, set uncertainty, or rank-interval reduction. A computed VOI order remains a prospective next artifact and would not rank people or languages.

5. Equal-treatment safeguards

The executable implementation rejects the following cases before any empirical rank is produced:

1. a structural-zero or insufficient-evidence row carrying a conflicting positive score;
2. repeated population source and denominator identities presented as additive cells, even when the estimates differ;
3. duplicate event identifiers, self-conditioning, or cycles;
4. marginal factors labeled as conditional without numerator, denominator, and parent-event evidence;
5. different need chains summed into one language target;
6. persons and learner-years placed in one competition without a versioned conversion;
7. a resource-gap scalar not derived from the exact verified audit bytes;
8. a compute value not reproduced from the closed registry and exact normalized inputs;
9. incomplete joint gain or compute draws used for a point rank;
10. qualitative beneficiary overlap presented as a quantified portfolio gain;
11. project status, local holdings, prior translation completion, sunk compute, national income, prestige, region, or demographic quotas entering the model; and
12. unresolved targets or missing evidence silently disappearing from the universe.

These controls create procedural equality, but procedural equality alone is insufficient. Target granularity, evidence availability, census definitions, and ecological proxies can still affect who is measurable. The study therefore publishes the ranked and unranked universes together and audits every macro/child, spoken/written, script, territory, and stage relation.

6. Results: inclusion first, then comparable orders

6.1 What the global sweep contains

The evidence freeze represents 1,077 exact, unresolved, regional, and intervention authority targets. Of these, 204 exact targets have source-authorized person denominators and enter the direct-person model; 2 targets enter a separate stage-opportunity order; and 871 remain explicitly unranked or contextual. Among the unranked rows, 775 lack a source-bounded person denominator. The public-safe score table leaves those population and score fields empty rather than displaying a synthetic country or world ceiling.

The high-consequence sentinel audit contains 145 checks: 136 have an exact target, 1 has only a broader base-language target, 6 have a context stratum but no exact edition target, and 2 remain missing. The two unresolved exact-target gaps are Algerian Amazigh varieties and Hmong varieties; neither receives a fabricated unified language identity or population. The context-only checks include major English and French education-language contexts that need exact comfort, stage, and resource endpoints before they can be compared.

This is a global gap map, not an assertion that all human languages have been observed. Its release condition is narrower and operational: no known high-consequence population is allowed to disappear merely because its available measure is incompatible with a speaker rank. Every such case must appear as a direct row, stage cohort, context/value-of-information row, structural portfolio, or named evidence gap.

6.2 Direct-person comparable order

The reference L-M order contains 204 source-authorized person-language targets. The table below shows the first ten; Appendix A prints all 100 leading rows. Of the Top 100, 89 have point constructs and 11 retain non-degenerate source intervals. A point constructed from a published count is not presented as a confidence interval; a range from a different construct is never attached as its low or high endpoint.

Rank	Exact target	Source population	Model median unmet need	Top-10 probability
1	Hindi (hi-Deva-IN)	322,230,097	119,286,966	100.0%

Rank	Exact target	Source population	Model median unmet need	Top-10 probability
2	Bahasa Indonesia (id-Latn-ID)	248,501,794	87,463,029	100.0%
3	Russian - Russian Federation (ru-Cyrl-RU)	134,319,233	48,028,115	100.0%
4	Punjabi (Pakistan standard) (pnb-Arab-PK)	88,915,544	38,667,896	100.0%
5	Bengali (India standard) (bn-Beng-IN)	96,177,835	35,118,380	98.0%
6	Marathi (mr-Deva-IN)	82,801,140	30,978,380	96.5%
7	Telugu (te-Telu-IN)	81,127,740	30,332,495	94.9%
8	Tamil (India standard) (ta-Taml-IN)	68,888,839	25,238,208	80.1%
9	Javanese (jv-Latn-ID)	68,044,660	24,717,315	75.4%
10	Central Thai (th-Thai-TH)	64,080,191	21,854,153	47.3%

These are model-conditional expected-need ranks, not directly observed global facts. Target-specific open-resource scarcity remains unmeasured for the scored universe, so all 206 direct-person and stage rows use the same 0.5 resource prior. Schooling-language gap uses the 0.4 common prior for 178 of 206 rows; academic-language non-overlap uses 0.4 for 201; and accessibility uses 0.5 for 204. Country learning/completion and delivery context provide more variation, but they remain ecological proxies where no target-specific cross-tab exists.

That evidence structure is visible in the sensitivity result. The Spearman correlation between model rank and the descending model-population point is 0.9989. Across four common-prior families and seven factor-removal/reference scenarios, nine targets remain in every Top 10: Hindi, Bahasa Indonesia, Russian - Russian Federation, Punjabi (Pakistan standard), Bengali (India standard), Marathi, Telugu, Tamil (India standard), Javanese. The union of possible Top-10 members adds Central Thai, Gujarati, Pashto (Pakistan curriculum pack). In the reference run Central Thai is tenth; under the scenario set the boundary can instead admit Gujarati or Pashto. Scenario rank correlations with the reference range from 0.9996 to 0.9999.

The correct funding interpretation is therefore two-part. The stable high ranks identify large source-bounded populations for which an open package could have high absolute reach. They do not prove that the same package, stage, or subject is missing in each language. Production should proceed with a target-specific resource and stage audit embedded in the same commission, while the value-of-information lane measures factors capable of changing the order.

Taiwan's official 2020 table reports current primary-or-secondary use by 21,090,192 resident nationals aged six or older for Mandarin, 18,728,839 for Taiwanese/Taigi, and 1,193,332 for Hakka. These appear at ranks 32, 33, and 67 as standalone interventions, but overlap and cannot be summed as 41,012,363 unique people; Traditional written Chinese is a different access endpoint (Directorate-General of Budget, Accounting and Statistics, 2020).

6.3 Standardized new-compute planning order

The second primary order divides the same model-conditional unmet-need burden by the standardized fresh-equivalent compute estimate for one fixed from-scratch package. Because no common package-effect estimate is available, this is a burden-per-compute planning proxy, not an estimate of causal access gained. It does not convert tokens to money, estimate a contractor budget, or claim that an end-to-end job has been observed. Every row in the present order uses a model-conditional script envelope; differences are therefore planning assumptions to be stress-tested, not measured production advantages.

Efficiency rank	Need rank	Exact target	Model median unmet-need equivalent per million FECU	Compute class
1	1	Hindi (hi-Deva-IN)	97,776,202	MODEL_CONDITIONAL_SCRIPT_ENVELOPE
2	2	Bahasa Indonesia (id-Latn-ID)	87,463,029	MODEL_CONDITIONAL_SCRIPT_ENVELOPE
3	3	Russian - Russian Federation (ru-Cyrl-RU)	44,470,477	MODEL_CONDITIONAL_SCRIPT_ENVELOPE
4	4	Punjabi (Pakistan standard) (pnb-Arab-PK)	33,919,207	MODEL_CONDITIONAL_SCRIPT_ENVELOPE
5	5	Bengali (India standard) (bn-Beng-IN)	28,094,704	MODEL_CONDITIONAL_SCRIPT_ENVELOPE
6	6	Marathi (mr-Deva-IN)	25,392,114	MODEL_CONDITIONAL_SCRIPT_ENVELOPE

Efficiency rank	Need rank	Exact target	Model median unmet-need equivalent per million FECU	Compute class
7	9	Javanese (jv-Latn-ID)	24,717,315	MODEL_CONDITIONAL_SCRIPT_ENVELOPE
8	7	Telugu (te-Telu-IN)	24,265,996	MODEL_CONDITIONAL_SCRIPT_ENVELOPE
9	8	Tamil (India standard) (ta-Taml-IN)	20,353,394	MODEL_CONDITIONAL_SCRIPT_ENVELOPE
10	10	Central Thai (th-Thai-TH)	18,520,469	MODEL_CONDITIONAL_SCRIPT_ENVELOPE

The efficiency and need orders are close because the current script-envelope estimates vary far less than the source-bounded population denominators. Javanese moves from need rank 9 to efficiency rank 7 under the Latin-script envelope, while Telugu and Tamil move down one position under their declared envelopes. This movement is useful for planning but is not evidence that one community's need matters more. The complete Top 100 efficiency order appears in Appendix A.2 and as a machine-readable CSV.

6.4 Stage opportunities: Chinese and Japanese

Some very large educational opportunities are observed as learner cohorts rather than whole-language readerships. They are not excluded; they are ordered in their own unit:

Stage rank	Exact opportunity	Cohort	Model median	Definition
1	Standard Written Chinese (zh-Hans-CN)	280,404,300	91,804,506	All formal-education enrolments reported by MOE in 2025; a stage-opportunity universe, not a count needing new Chinese material.
2	Standard Japanese (ja-Jpan-JP)	2,922,969	1,291,650	2,645,837 undergraduates plus 277,132 graduate/professional-degree students.

The Mainland Chinese row makes Simplified Standard Written Chinese one of the largest immediately visible educational opportunities in the study. The relevant first commissions are not a generic claim that China lacks textbooks. They are open and adaptable advanced STEM, research methods, reproducibility, open science, accessible semantic formats, vocational/adult routes, and targeted rural, migrant, disability, or field-specific gaps. Spoken Putonghua and written literacy remain separate constructs.

Japan is not a negative control for educational worth. Its strong general educational system narrows the likely marginal package: research and postgraduate reading/writing, open science and reproducibility, accessible advanced STEM, specialist professional material, international terminology bridges, and exact support cohorts. The 2,922,969 opportunity row is tertiary enrolment, not the Japanese-speaking population; other directly recorded cohorts remain in the evidence register. A smaller stage denominator can still support high-leverage research or accessibility work.

6.5 The funding portfolio

No responsible allocation can be made from the direct-person Top 100 alone. The release therefore exposes four coequal decision lanes:

Decision lane	Rows	What can be compared	Immediate use
1. Direct-person comparable	100	Source-authorized person-language rows under the same model	High-reach package and evidence prioritization
2. Stage opportunity	2	Direct learner/sector cohorts with each other	Stage-specific commissioning without inventing whole-language readers
3. Mandatory context and value of information	19	No cross-row unmet-need rank	Keep high-scale and undermeasured populations visible; collect allocation-changing evidence
4. Structural access portfolio	65	Only within exact compatible units	Named sign languages, accessibility/offline modes, host-specific displacement continuity, and bounded bridge experiments

For an initial high-reach compute programme, the first direct-person slate is the nine scenario-robust targets, with Thai, Gujarati, and Pashto treated as the empirically unresolved tenth-position set rather than separated by a cosmetic tie-break. This is a priority for fitted open packages, not a recommendation to translate one identical book into each language. Simplified written Chinese should be commissioned simultaneously through its stage lane because the 280.4-million formal-education cohort is too large to disappear from a speaker-only table. Japan receives a narrower advanced/research/accessibility programme. Named sign-language and accessibility modes receive a minimum-service lane even where population measurement is weak. Displaced learners receive host-specific source-language continuity, host-language acquisition, credential, stage, and offline packages. Bridge and interlanguage work is funded as measured comprehension experiments and shared production infrastructure, not as unverified population substitution.

6.6 What educational material is needed

The programme must distinguish early childhood, primary, secondary, vocational/adult, undergraduate, postgraduate, and research access. "Education" includes frontier monographs, research methods, standards, software and data documentation, and specialist professional learning; those resources serve smaller cohorts but can have high institutional leverage. The evidence-based first and second package classes for leading routes are:

Population or route	Stage order	First package classes	Then expand to	Delivery
Standard Hindi	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate -> research	foundation/catch-up; mathematics/science; health; teacher education	computing/data; finance; TVET; degree spines; research methods; advanced material	offline/print/mobile; solutions; bilingual academic terminology
Standard Indonesian	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate -> research -> early-childhood	mastery literacy/numeracy; complete mathematics/science; health; climate/disaster; teacher education	TVET; computing/data; finance/management; degree spines; research methods; advanced research	offline HTML/EPUB/PDF; small chapters; audio; print; solutions; local code/data
Russian	research -> postgraduate -> undergraduate -> accessibility	open science; current methods; accessible advanced STEM	frontier monographs; computing/data; specialist professional updates	source/HTML/MathML; bilingual metadata; offline code/data
Punjabi; Shahmukhi	early-childhood -> primary -> lower-secondary -> vocational-adult	foundation; Urdu bridge; health	agriculture; climate/disaster; enterprise finance; secondary scaffold	audio/print; bilingual Punjabi-Urdu; offline; locale-tagged

Population or route	Stage order	First package classes	Then expand to	Delivery
Bangla	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate -> research -> early-childhood	foundation/catch-up; mathematics/science; health; teacher education; climate/disaster	garment/manufacturing TVET; agriculture; finance; computing/data; degree spines; research methods	offline/mobile/print; chapter downloads; audio; full solutions; local practicals
Marathi	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate	foundation/catch-up; mathematics/science; health	industry/TVET; finance/management; computing; degree spines; research methods	offline/print/mobile; worked solutions; bilingual terminology
Telugu	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate	foundation/catch-up; mathematics/science; health	computing/data; agriculture; industry TVET; degree spines; research methods	offline/print/mobile; code/data local; bilingual academic terms
Tamil	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate -> research	foundation/catch-up; mathematics/science; health	industry/TVET; computing/data; degree and advanced research spines	offline/print/mobile; MathML; bilingual metadata
Javanese; Latin core	early-childhood -> primary -> lower-secondary -> vocational-adult	oral/literacy/numeracy foundation; Indonesian bridge; health; disaster	selected secondary STEM terminology; agriculture; business/finance; teacher support	audio plus print; dialect-tagged; bilingual glossary; small offline units
Standard Thai	lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate -> research	mathematics/science progression; health; digital/financial capability	TVET; computing/data; degree and research methods; open science	offline/source/print; worked solutions; bilingual academic terms
Standard Mandarin; Simplified Chinese	postgraduate -> research -> undergraduate -> vocational-adult -> upper-secondary	research methods; reproducibility; open science; accessible STEM reference	advanced mathematics and science; computing/data/AI; health professional; selected frontier monographs	searchable HTML; source; MathML; EPUB/PDF; local code/data; bilingual search terms
Japanese	research -> postgraduate -> undergraduate -> accessibility	research reading/writing; open science; reproducibility; accessible advanced STEM	frontier mathematics/science; specialist professional material; international terminology bridges	source-available HTML/PDF/EPUB; MathML; bilingual metadata; local code/data

For Bengali in India, the source-authorized population row is exact but the present stage-subject map is not country-specific enough to borrow Bangladesh's learning and climate profile. The shared Bangla semantic source can lower production cost, while India and Bangladesh retain separate curriculum, terminology, examples, standards, and outcome evidence. Taiwan Mandarin, Taigi, and Hakka likewise need separate spoken/written and literacy endpoints before one package is treated as coverage for another.

Across the wider Top 100, the recurring high-utility bundles are: foundational literacy/numeracy/science with cumulative practice and complete solutions; planned strongest-language-to-school-language bridges; secondary STEM and prerequisite recovery; health; agriculture, water, climate, and disaster resilience; computing and data; household and enterprise finance; management and cooperative governance; teacher/facilitator education; vocational safety and measurement; undergraduate gateway sequences; research methods and open science; and advanced specialist corpora. Appendix A identifies who is in the comparable orders; Appendix E prints the 63-row stage-subject recommendation register used for the leading and structurally important routes.

6.7 Open and autonomous delivery

Open licensing is an operational requirement: newly commissioned materials must be lawful to download, retain, copy, adapt, translate, and redistribute (UNESCO, 2019). Autonomous packages must not assume a teacher, continuous connectivity, a laptop, or a paid service; they require prerequisite maps, diagnostics, explanation, practice, reasoned answers, mastery checks, and a route forward. Delivery branches remain independent: semantic HTML with MathML, accessible EPUB/PDF, low-data audio with transcripts, captioned or named-sign-language video, plain-language companions, and small offline bundles. OpenStax's 73-title catalogue is a strong upper-secondary/introductory-tertiary baseline, not a complete world curriculum; the 45 complementary records cover early learning, computing, health, agriculture, climate, research methods, open science, and advanced reference (OpenStax, n.d.).

7. Material priorities

The educational package map treats OpenStax as a transparent, exercise-rich baseline rather than a definition of need. Its catalogue is strongest for upper-secondary and introductory tertiary subjects. Complementary open resources are required for early learning, teacher education, computing and data, health, agriculture and climate, vocational education, research methods, open science, and advanced reference.

Across high-learning-poverty and high-language-mismatch settings, priority often begins with first-language or strongest-language literacy, numeracy, science foundations, diagnosis, cumulative practice, teacher or caregiver guidance, and offline/audio/print delivery. Large secondary and adult populations add prerequisite recovery, applied mathematics, computing, health, finance, management, agriculture, climate and disaster resilience, and vocational safety or measurement. Tertiary systems require complete gateway sequences, not isolated textbook chapters. Postgraduate and research users require methods, statistics, reproducibility, literature navigation, scholarly writing, terminology, open science, and advanced specialist texts.

Language-specific recommendations follow local evidence. Examples include climate-smart agriculture for rural and climate-exposed communities; bookkeeping, finance, marketing, and management for informal enterprises; health literacy for rural, Indigenous, refugee, and migrant communities; bilingual teacher education where instruction transitions between home and academic languages; and research-navigation bridges in large languages whose scholars face substantial English-language burdens.

7.1 A stage-complete educational ladder

Educational access is not synonymous with primary school access. The intervention census therefore distinguishes eight stages. A language can have strong compulsory-school provision and still have a serious university, professional, or research-language gap; conversely, a translated specialist monograph cannot substitute for missing foundational literacy or numeracy. Stage populations are kept disjoint in the quantitative model, and recommendations are allowed to differ by stage within the same language.

Stage	First-line open package	Additional package	Autonomous and offline form
Early childhood	Strong-language conversation, stories, vocabulary, counting, comparison, shape, pattern, hygiene, nutrition, and safety	Play-based observation, prediction, local environment, and social-emotional routines	One-page caregiver cues, picture-story activities, basic-phone audio, and everyday-object alternatives; no literate adult is presumed
Primary	Reading and writing in the learner's strongest route, an explicit bridge to the later instructional language, arithmetic, fractions, measurement, geometry, inquiry science, health/WASH, and local climate or hazard basics	Unplugged computing, digital safety, household numeracy and money, food systems, and environmental observation	Short mastery units, worked examples, complete explanatory answers, cumulative review, print-first completeness, and audio/visual alternatives
Lower secondary	Proportional reasoning and algebra, statistics, integrated biology/chemistry/physics, reading-to-learn, health and sexual/reproductive health, and climate/disaster readiness	Programming and data, cybersecurity, personal finance, design and making, and evaluation of claims and evidence	Worked-example/problem pairs, low-cost practicals, answer keys with reasoning, downloadable simulations, and print fallbacks
Upper secondary	Pathway mathematics and statistics, biology, chemistry, physics, technical and academic writing, digital security, health, and water/energy/climate systems	Calculus, computer science, data science, economics, accounting, entrepreneurship, and supervised or autonomous research projects	Modular academic and technical routes with shared prerequisite maps, offline code/data notebooks, complete solutions, and exam-independent mastery checks
Vocational and adult	Functional literacy and numeracy, occupational safety, health, financial and digital capability, and job-specific competencies	Entrepreneurship, bookkeeping, pricing, inventory, marketing, project and cooperative management, climate-smart production, and recognition-of-prior-learning packs	Demonstration-practice-performance sequences, illustrated or audio guides, local units and tools, short stackable modules, and explicit update dates
Undergraduate	First-year mathematics, statistics, writing and study bridges; core discipline texts; computing/data; and research literacy	Complete open degree spines in mathematics, science, engineering, health, agriculture, economics, management, education, and social science	Downloadable books with exercises and solutions, executable local examples, bilingual terminology, prerequisite maps, and accessible HTML/EPUB/PDF

Stage	First-line open package	Additional package	Autonomous and offline form
Postgraduate	Advanced theory, quantitative and qualitative methods, scholarly reading and writing, research ethics, data management, and reproducible workflows	Specialist monographs, frontier methods, research software, project/grant management, and disciplinary literature maps	Source-available books, datasets and code, problem sets, stable citations, searchable terminology, and versioned updates
Research and professional	Current reviews, standards, methods, software and data documentation, open science, professional safety, and decision tools	Frontier monographs and research corpora selected by field need rather than school-age population	Searchable semantic HTML, source files, structured data, formula-safe PDF and MathML, downloadable reference bundles, and explicit update paths

This ladder prevents two opposite allocation errors. It does not send advanced research texts to populations whose binding constraint is foundational access; it also does not treat researchers, professionals, instructors, or postgraduate learners as already served merely because compulsory schooling exists in their language. Research is an educational process, and translation of advanced theory, methods, standards, and research navigation can have high system leverage even when its direct audience is smaller.

7.2 Subject bundles and population-specific utility

The common canon is not mathematics-only. Mathematics is retained as a particularly reusable and auditable domain - formal notation, worked problems, executable checks, and answer verification lower some production and validation costs - but the need analysis covers the wider educational system.

Bundle	Minimum coherent content	Conditions producing outsized marginal value
Literacy, mathematics, and statistics	Strong-language literacy; planned bridge literacy; numeracy through algebra, calculus, probability, statistics, modelling, and formal reasoning	Learning-poverty settings, language-of-instruction mismatch, secondary re-entry, vocational transitions, first-year quantitative bottlenecks, and research-method deficits
Biology, chemistry, physics, and earth/environmental science	Conceptual sequences, quantitative problem solving, low-cost practicals, laboratory and field safety, measurement, data interpretation, and local examples	Weak secondary progression; health, agriculture, engineering and climate pathways; laboratories with limited equipment; first-year remediation
Health	Prevention, maternal and child health, infectious and noncommunicable disease, mental health, sexual and reproductive health, medicine safety, risk interpretation, and service navigation	Rural and remote communities, Indigenous and minority-language users, displaced populations, community health workers, and low-health-literacy adult cohorts
Agriculture, water, climate, and disaster resilience	Soil and water, crops and livestock, pest management, weather and market information, adaptation, biodiversity, coastal and flood risk, and disaster planning	Smallholders, pastoralists, fishers, coastal/delta/island populations, climate migrants, rural women, and disaster-exposed regions
Computing and data	Device and information literacy, privacy, scams and cybersecurity, spreadsheets, programming, databases, data cleaning and analysis, visualization, AI limits, provenance, and evaluation	School-to-work transitions, low computer ownership, mobile-first users, public-service access, small enterprises, tertiary study, and research communities dependent on English-language tooling
Economics, finance, and management	Household finance, consumer protection, digital payments, bookkeeping, pricing, inventory, marketing, entrepreneurship, cooperative governance, operations, and project management	Underbanked adults, informal enterprises, farmers and fishers, women-owned and rural businesses, vocational learners, and local public or nonprofit organizations
Teacher and facilitator education	Subject knowledge, structured pedagogy, diagnosis and targeted instruction, multilingual teaching, inclusion, accessibility, low-cost practicals, and assessment interpretation	Multilingual systems, conflict or displacement, underqualified-teacher systems, home-to-school language transitions, and rapid expansion of secondary or tertiary provision
Research methods and open science	Research design, measurement, probability and statistics, causal and qualitative methods, coding, data/software management, ethics, reproducibility, citation, peer review, open licensing, and scientific communication	Every research community not fully comfortable in the dominant international literature; growing tertiary systems; local health, agriculture, climate, engineering, and policy research
Advanced research	Complete specialist monographs, rigorous problem sets, current reviews, standards, research software documentation, and frontier corpora	Concentrated research and instructor communities, including languages with strong school systems but substantial English or other academic-lingua-franca burdens

The ordered package for a target is selected from measured constraints, not from national income or an assumption that every population wants the same book. High foundational learning gaps move complete literacy, numeracy, science, and teacher/facilitator packages forward. Large adult livelihood cohorts can move finance, management, health, agriculture, climate, safety, or computing packages forward. Large tertiary and research cohorts with thin open native-language sequences can move gateway, methods, and advanced work forward even when school outcomes are strong.

7.3 Open, autonomous, and accessible by construction

For the intended access outcome, open availability is a functional requirement. A package that cannot lawfully be downloaded, retained, copied, adapted, translated, and redistributed cannot support a global language programme on equal terms. "Free to read," "openly licensed," "complete for self-study," "available offline," and "accessible" are therefore separate audited properties; none is inferred from another.

The current canon audit makes the supply boundary concrete. The language-neutral canon contains 118 records: 73 OpenStax single-course textbooks and 45 complementary open or reference resources. The OpenStax records are concentrated at upper-secondary and introductory tertiary levels (65 records at ISCED 5 and 64 at ISCED 6); it has no level-0, level-1, level-7, or level-8 record and only one level-2 record. The eight Open Logic records form a narrow formal-logic and research pathway rather than a general curriculum. Domain and ISCED memberships are overlapping descriptive classifications generated from catalogue/title fields, not unique-title counts, importance weights, or proof of target-language coverage. These counts are catalogue records, not proof that a target language has a usable translated route.

In the exact target-stage-canon sweep, 148 target profiles were crossed with seven stages (E0-E6) and five controlled subject canons, producing 5,180 cells. Only 32 cells were `RESOURCE_FOUND`; all 32 are Hindi or Urdu NCERT/ePathshala routes at selected stages and canons. They establish free access at those endpoints but fail the audited open-adaptation/redistribution tests. The remaining 5,148 cells are `SEARCH_UNRESOLVED` (4,852 F0 and 296 F1), not evidence that no resource exists. No numeric resource-scarcity value is imputed from either state. The matrix uses `lang:zh-Hans-CX` for its Chinese route while the population target is `zh-Hans-CN`, and its `lang:id-Latn-ID` rows are not the separate national SIBI evidence route; these are explicit crosswalk gaps, not evidence of absence or proof of coverage. The exact-cell matrix and validation receipt are therefore part of the paper's evidence, not a hidden assumption behind its ranking.

The supply-fit implications are staged rather than language-blind:

Need band	Strongest current open components	What remains an unmet commissioning task
E0-E1 foundation and early numeracy	African Storybook, Global Digital Library early-reading/early-math candidates	Complete diagnostic/mastery spine, planned language bridge, answers, and verified accessibility/offline route
E1-E3 primary/lower-secondary STEM and device-light computing	Siyavula mathematics/science, CS Unplugged, PhET, selected OpenStax supplements	Whole-school sequence, local-language examples, complete feedback, and rights/format checks
E3-E6 quantitative STEM and formal reasoning	OpenStax algebra/statistics/calculus/science, OpenIntro, Active Calculus, forallx/Open Logic	Prerequisite map, localized terminology/examples, complete solutions, and target-language evidence
E4-E7 health, agriculture, climate, and TVET	Open RN, OpenStax nursing, soil/environmental texts, FAO manuals, Carpentries/Turing Way	Community-language self-study spine, clinical/jurisdictional updating, occupational practice and assessment
E5-E8 advanced research	Open Logic, selected MIT OCW, Stacks Project, research-methods and open-science resources	Broad specialist corpus, multilingual research navigation, source completeness, and accessible semantic delivery

The mapping is a conditional supply inventory. It does not claim that a catalogue title is available in a target language, that a free download is adaptable, or that a component forms a complete autonomous curriculum. A many-to-many stage/subject crosswalk is retained because the needs map uses 63 material-priority rows while the exact-cell canon uses 5,180 target-stage-canon rows.

Every core learner package should include a prerequisite map and entry diagnostic; short explanations with explicit worked models; guided and independent practice; complete answers with reasoning; mastery checks and cumulative review; a glossary connecting the learner's strong language, national bridge language, and international search terminology; low-bandwidth downloadable source; accessibility alternatives carrying the same essential content; and a route to the next stage. A teacher or facilitator edition can add lesson routines, diagnostic guidance, common misconceptions, multilingual pedagogy, and low-cost practicals, but teacher availability is not a prerequisite for learner access.

Delivery is evaluated as a set of independent routes: semantic HTML and MathML for screen-reader and reflow use; EPUB; tagged PDF; print-ready files; captioned or signed video; audio/DAISY where the content permits it; plain-language companions; and small, mirrorable offline bundles for phone, local server, USB, or memory card. Essential mathematics, diagrams, or instructions may not be trapped in unlabelled images. Code and data examples must run without a mandatory paid service. Health, agriculture, finance, law, safety, and other time-sensitive material requires visible jurisdiction, version date, and maintenance route.

8. Interlanguages and bridge designs

An interlanguage can be valuable below perfect comprehension. The relevant outcome is functional increase in comprehension at a stated stage and genre, net of training burden and overlap with existing editions. Its name or family affiliation supplies no population evidence.

Mathematics deserves a domain-specific treatment. Formal notation can carry relations for readers who already possess sufficient mathematical background, which may make symbol-dense worked examples or formulas more accessible across related languages than ordinary prose. It does not remove the language burden of definitions, hypotheses, quantifiers, proof structure, explanations, word problems, or pedagogy. The study therefore distinguishes worked examples, definitions, proof prose, word problems, and research reference use rather than assigning one comprehension multiplier to "mathematics."

The literature supports a conditional bridge advantage rather than either blanket dismissal or blanket coverage. Formula-plus-text or formula-plus-graphic representations can help when their roles and notation are familiar, while symbol-bearing conceptual text can also increase reading difficulty. Exact numerical knowledge can remain tied to the language in which it was learned, and language proficiency predicts contextual or word-problem performance more strongly than bare computation (Österholm, 2006; Ott et al., 2018; Malone et al., 2020; Spelke & Tsivkin, 2001; Hahn et al., 2019; Hickendorff, 2013; Trakulphadetkrai et al., 2020). The plausible breakpoint is therefore an interaction among prior mathematical knowledge, notation familiarity, technical lexicon, prose density, learner stage, and task purpose. Advanced readers consulting a familiar argument can exploit partial prose recovery more effectively than beginners learning a definition or solving a contextual problem.

No reviewed profile currently has a representative estimate of mathematical learning gained through an interlanguage or receptive bridge. General cloze, word recognition, short-text comprehension, training studies, and technical-lexicon comparisons justify experiments and production sharing, but they do not identify proof comprehension, transfer, or family-wide reach. The current bridge lane consequently distinguishes: (1) shared-source parallel natural standards that can be deployed while preserving labels, scripts, and terminology; (2) natural receptive pairs and learned intercomprehension strategies warranting direct mathematics experiments; (3) learned regional or international languages serving measured proficient users; and (4) speculative constructed codes that remain hypotheses. Missing mathematics-specific gain remains unidentified rather than being assigned zero.

The bridge matrix records exact communities, script and orthography, comprehension without prior study where measured, learned exposure, shared-source standardization, overlap with English or native editions, displaced or diaspora use, non-served groups, and uncertainty. It includes evidence and hypotheses for Interslavic, Romance intercomprehension and constructed Romance designs, Turkic parallel or bridge designs, Persian-Dari-Tajik, Hindi-Urdu, Scandinavian receptive multilingualism, Kiswahili, Modern Standard Arabic, West African Pidgin varieties, Esperanto, International Sign, and possible Germanic profiles. Constructed Inter-Germanic remains a research hypothesis; a pan-Germanic register is not assumed to serve Low German, Plautdietsch, Pennsylvania Dutch, Hutterisch, Frisian, Afrikaans, Yiddish, Scots, Faroese, or other communities without direct evidence.

Production efficiency and beneficiary reach remain separate. A shared semantic source, terminology layer, or script-conversion pipeline can reduce compute for several editions even when no reader understands the common source directly. Conversely, demonstrated no-study comprehension can increase reach but may still require separate scripts, terminology, examples, or curriculum overlays.

9. Reproducibility and release design

The public package separates authority, evidence, modeling, and decision outputs. The authority contains 1,077 targets. The evidence authorization ledger has 850 records and admits 259 population observations to 204 direct-person targets; three stage observations collapse to 2 stage targets. The source-to-score crosswalk contains 850 rows. The public-safe score table contains 1,077 rows and suppresses non-empirical population values for 871 unranked targets.

The direct-person source-trace audit passes all target, authorization, source, bound-normalization, forbidden-predictor, and rank-sequence checks. The territory-binding audit checks 921 valid BCP-47 region rows and 87 unresolved identities with zero mismatches. The portfolio audit contains 186 rows across the direct, stage, context/value-of-information, structural, displacement, and bridge lanes. The inclusion audit reports `PASS_WITH_EXPLICIT_EVIDENCE_GAPS`: that status is an honest description of a versioned global search, not a claim that every language has a measured denominator.

Key frozen artifacts are:

Artifact	Purpose	SHA-256
structured/GLOBAL_TARGET_SCORE_TABLE_PUBLIC_v3.csv	Public-safe all-target score and disposition table	08845507E3EB0B1963249B5227A657913503093BA00B823F2D383FA19ECE0313
structured/TOP100_NEEDS_ONLY_v3.csv	Full direct-person Top 100	B97EF9CD8322226C20F5BEDE8BCD8E7E19BD7995C2D2DB7169D5BBD40E38D183
structured/GLOBAL_FUNDING_PORTFOLIO_v1.csv	Multi-lane allocation portfolio	126114DE07AE1A86471E448AB76C9D6AE409656C89930414CDFA87E4A91CEDB3
structured/GLOBAL_INCLUSION_DISPOSITION_REGISTER_v1.csv	Full authority/context/structural/bridge/sentinel disposition	96EBED123BC08AEC801FAC87508506D04A3D96A820B1FE1026F28FAB88DFDB67
structured/SOURCE_TO_SCORE_CROSSWALK_v1.csv	Source-record disposition crosswalk	9ED049D05D47A0DED2842E9A74966A939FEDB2F7DA7A60D3AA9D27611AAB719A

Artifact	Purpose	SHA-256
structured/RANK_SENSITIVITY_SUMMARY_v1.csv	Prior and factor-removal rank stability	2EB2DCDD66AC0055CFABEA900A4DA76F47F765B4415A4688B79257BF4280CE3D
structured/TOP100_COMPUTE_EFFICIENCY_v3.csv	Standardized new-compute Top 100 planning order	C98A2C2CAF1ABF972EB277CE8656C2C5E5CCDB698AABE4039D918F1595B41E11
agent_reports/STAGE_SUBJECT_NEEDS_MAPPING.csv	Population-stage-subject recommendations	EC5E15932D9EE33CC4ACCD25F6B85C81D992660195C7FB753849437964C98132
agent_reports/INTERLANGUAGE_STRUCTURED_EVIDENCE_MATRIX.csv	Bridge/interlanguage evidence and non-served groups	BD5D2128A69A042069D74AD27906FA9A08A27A2E8A9F8EC93845BA8827E118CF

The executable model retains fixed target-set equality, source and row hashes, a forbidden-predictor scan, deterministic SHA-256-addressed draws, L-ID partial-identification bounds, four prior families, seven factor-removal scenarios, and explicit script-envelope compute assumptions. The compute comparison is a fresh-equivalent planning unit, not a price conversion. Local holdings, prior project completion, sunk compute, personal preferences, national income, prestige, and demographic quotas are absent from the model and the portfolio builder.

The canonical public repository is https://github.com/KokunoYumeto/allocating-ai-translation-compute-for-educational-access. The release includes this paper as PDF, DOCX, Markdown, and readable offline HTML together with the machine-readable tables, sources, scripts, and sanitized QA receipts. Public-byte readback is required after both GitHub and Zenodo publication.

10. Limitations

Academic-language comprehension, community-specific learning outcomes, open-resource coverage, overlapping access constraints, production feasibility, mathematics-specific interlanguage performance, and causal package effects remain incompletely measured. Official status and exposure do not establish proficiency. These gaps govern uncertainty, rankability, and evidence acquisition; they do not justify convenient proxies or excluding underdocumented populations. Every claim retains its denominator, and unsupported quantities remain missing.

Appendix A. Full direct-person and standardized-compute orders

A.1 Direct-person Top 100

The table is complete for the comparable reference need order. Population entries retain points or source intervals; model medians are rounded display values. Exact definitions, sources, URLs, hashes, uncertainty fields, and compute estimates are in `structured/GLOBAL_FUNDING_PORTFOLIO_v1.csv` and `structured/TOP100_NEEDS_ONLY_v3.csv`. Rows are standalone interventions and must not be summed without a disjoint-person crosswalk.

Rank	Exact target	Source population	Reference year	Model median	Top-100 probability
1	Hindi (hi-Deva-IN)	322,230,097	2011	119,286,966	100.0%
2	Bahasa Indonesia (id-Latn-ID)	248,501,794	2022	87,463,029	100.0%
3	Russian - Russian Federation (ru-Cyrl-RU)	134,319,233	2020 Census; reference date 2021-10-01	48,028,115	100.0%
4	Punjabi (Pakistan standard) (pnb-Arab-PK)	88,915,544	2023	38,667,896	100.0%
5	Bengali (India standard) (bn-Beng-IN)	96,177,835	2011	35,118,380	100.0%
6	Marathi (mr-Deva-IN)	82,801,140	2011	30,978,380	100.0%
7	Telugu (te-Telu-IN)	81,127,740	2011	30,332,495	100.0%
8	Tamil (India standard) (ta-Taml-IN)	68,888,839	2011	25,238,208	100.0%
9	Javanese (jv-Latn-ID)	68,044,660	2010	24,717,315	100.0%
10	Central Thai (th-Thai-TH)	64,080,191	2010-09-01	21,854,153	100.0%
11	Gujarati (gu-Gujr-IN)	55,036,204	2011	20,261,192	100.0%
12	Pashto (Pakistan curriculum pack) (ps-Arab-PK)	43,633,946	2023	19,363,224	100.0%
13	Bhojpuri (bho-Deva-IN)	50,579,447	2011	18,937,757	100.0%
14	Urdu (ur-Arab-IN)	50,725,762	2011	18,746,774	100.0%

Rank	Exact target	Source population	Reference year	Model median	Top-100 probability
15	Burmese (my-Mymr-MM)	36,000,000-41,000,000	2022	17,171,729	100.0%
16	Uzbek (uz-Latn-UZ)	35,700,000	2026	16,913,981	100.0%
17	Kannada (kn-Knda-IN)	43,506,272	2011	16,061,994	100.0%
18	Sindhi (Pakistan standard) (sd-Arab-PK)	34,401,564	2023	15,421,635	100.0%
19	Moroccan Arabic (ary-Arab-MA)	34,000,000 [33,000,000-35,000,000]	2024	14,878,712	100.0%
20	Spanish (es-Latn-US)	43,369,734	2023	13,908,336	100.0%
21	Malayalam (ml-Mlym-IN)	34,776,533	2011	13,036,703	100.0%
22	Saraiki (skr-Arab-PK)	28,849,579	2023	12,849,258	100.0%
23	Oromo (om-Latn-ET)	24,930,424	2007	12,790,119	100.0%
24	Odia (or-Orya-IN)	34,059,266	2011	12,732,595	100.0%
25	Standard Yoruba (yo-Latn-NG)	30,000,000 [28,000,000-32,000,000]	2026	12,548,589	100.0%
26	Sundanese (su-Latn-ID)	32,412,752	2010	11,691,959	100.0%
27	Standard (mg-Latn-MG)	23,484,462 [23,472,708-23,496,217]	2018	11,558,560	100.0%
28	Eastern Punjabi (pa-Guru-IN)	31,144,095	2011	11,321,067	100.0%
29	Amharic (am-Ethi-ET)	21,634,396	2007	10,832,145	100.0%
30	Igbo (ig-Latn-NG)	25,000,000 [23,000,000-27,000,000]	2026	10,300,842	100.0%
31	Urdu (ur-Arab-PK)	22,249,307	2023	9,880,681	100.0%
32	Taiwan Mandarin (Guoyu) (cmn-Hant-TW)	21,090,192	2020-11-08	9,109,832	100.0%
33	Taiwanese Taigi - Southern Min (nan-Hant-TW)	18,728,839	2020-11-08	8,049,853	100.0%
34	Kurmanji Kurdish (kmr-Latn-001)	15,000,000-20,000,000	2025	7,458,209	100.0%
35	Khmer (km-Khmr-KH)	14,893,134	2019	6,425,051	100.0%
36	Chhattisgarhi (hne-Deva-IN)	16,245,190	2011	6,149,264	100.0%

Rank	Exact target	Source population	Reference year	Model median	Top-100 probability
37	isiZulu (zu-Latn-ZA)	14,613,202	2022	6,036,704	100.0%
38	Nepali (Nepal standard) (ne-Deva-NP)	13,084,457	2021	5,609,219	100.0%
39	Assamese (as-Beng-IN)	14,816,414	2011	5,547,295	100.0%
40	Kazakh (kk-Cyrl-KZ)	12,670,407	2021	5,544,880	100.0%
41	Maithili (India) (mai-Deva-IN)	13,353,347	2011	4,959,628	100.0%
42	Magahi (mag-Deva-IN)	12,706,825	2011	4,708,245	100.0%
43	Catalan (ca-Latn-ES)	10,000,000 [9,000,000-11,000,000]	2026	4,307,845	100.0%
44	isiXhosa (xh-Latn-ZA)	9,786,928	2022	3,980,641	100.0%
45	Standard Kinyarwanda (rw-Latn-RW)	6,490,743 [6,486,597-6,494,888]	2022	3,073,509	100.0%
46	Marwari (mwr-Deva-IN)	7,831,749	2011	2,977,338	100.0%
47	Madurese (mad-Latn-ID)	7,743,533	2010	2,820,241	100.0%
48	Cantonese (Yue) (yue-Hant-HK)	6,328,947	2021	2,748,977	100.0%
49	Turkmen (tk-Latn-TM)	6,297,965	2022	2,657,005	100.0%
50	Standard Somali (so-Latn-ET)	4,609,274	2007	2,569,049	100.0%
51	Sorani Kurdish (ckb-Arab-001)	6,000,000-8,000,000	2016	2,517,345	100.0%
52	Kyrgyz (ky-Cyrl-KG)	5,360,056	2022	2,292,964	100.0%
53	Tigrinya (ti-Ethi-ET)	4,324,933	2007	2,180,833	100.0%
54	Paraguayan Guarani (gug-Latn-PY)	3,792,294	2025	1,629,362	100.0%
55	Georgian (ka-Geor-GE)	3,343,987	2024	1,601,359	100.0%
56	Minangkabau (min-Latn-ID)	4,230,000	2010	1,537,983	100.0%
57	Awadhi (India) (awa-Deva-IN)	3,850,906	2011	1,438,774	100.0%
58	Maithili (Nepal) (mai-Deva-NP)	3,222,389	2021	1,397,627	100.0%
59	Armenian (hy-Armn-AM)	2,840,631	2022	1,356,593	100.0%
60	Banjarese (bjn-Latn-ID)	3,650,000	2010	1,308,737	100.0%

Rank	Exact target	Source population	Reference year	Model median	Top-100 probability
61	Buginese (bug-Latn-ID)	3,510,000	2010	1,266,450	100.0%
62	Brahvi (brh-Arab-PK)	2,778,670	2023	1,198,792	100.0%
63	Nepali (India jurisdiction pack) (ne-Deva-IN)	2,925,796	2011	1,106,714	100.0%
64	Bhojpuri (Nepal) (bho-Deva-NP)	1,820,795	2021	772,107	100.0%
65	Guinea-Bissau Kriol (pov-Latn-GW)	1,303,743	2009	561,062	100.0%
66	Q’eqchi’ (kek-GT)	1,127,387	2018	520,136	100.0%
67	Taiwan Hakka (hak-Hant-TW)	1,193,332	2020-11-08	519,046	100.0%
68	K’iche’ (quc-GT)	1,054,818	2018	499,097	100.0%
69	Kreol Morisien (mfe-Latn-MU)	1,002,135	2022	425,209	100.0%
70	Basque (eu-Latn-ES)	936,812	2021	407,030	100.0%
71	Awadhi (Nepal) (awa-Deva-NP)	864,276	2021	371,819	100.0%
72	Mam (mam-GT)	590,641	2018	280,069	100.0%
73	Maayat’aan (yua-MX)	774,755	2020	269,577	100.0%
74	Shona (sn-Latn-ZA)	716,967	2022	264,793	100.0%
75	Standard Moroccan Amazigh (zgh-Tfng-MA)	542,781	2014	253,700	100.0%
76	Tseltal (tzh-MX)	589,144	2020	201,861	100.0%
77	Bolivian Aymara (ayr-BO)	457,358	2024	197,684	100.0%
78	Kaqchikel (cak-GT)	411,089	2018	196,634	100.0%
79	Welsh (cy-Latn-GB)	538,300	2021	195,519	100.0%
80	Aymara in Peru (ayr-PE)	444,389	2017	193,950	100.0%
81	Tsotsil (tzo-MX)	550,274	2020	190,213	100.0%
82	Russian mother-tongue population in Estonia (ru-Cyrl-EE)	379,210	2021	176,503	100.0%
83	Rwanda Kiswahili (swh-Latn-RW)	331,583 [327,438-335,729]	2022	159,651	100.0%
84	Tetun Prasa (tet-Latn-TL)	361,027	2015	157,971	100.0%

Rank	Exact target	Source population	Reference year	Model median	Top-100 probability
85	Magahi (mag-Deva-NP)	230,117	2021	98,701	100.0%
86	Maori (mi-Latn-NZ)	213,849	2023	93,581	100.0%
87	Belize Kriol (bzj-BZ)	180,792	2022	86,924	100.0%
88	Chol (ctu-MX)	254,715	2020	85,943	100.0%
89	Mambai (mgm-Latn-TL)	195,778	2015	81,855	100.0%
90	Q'anjob'al (kjb-GT)	166,261	2018	70,940	98.4%
91	Mexicano de la Huasteca hidalguense (nah-huasteca-hidalguense-Latn-MX)	195,770	2000	69,358	98.8%
92	Poqomchi' (poh-GT)	133,074	2018	56,620	94.1%
93	Makasae (mkz-Latn-TL)	123,840	2015	53,586	88.7%
94	Achi (acr-GT)	124,338	2018	52,961	90.6%
95	Diné Bizaad (nav-US)	161,174	2017-2021	51,686	89.8%
96	Ixil (ixl-GT)	114,997	2018	50,776	86.7%
97	Curaçao Papiamentu (pap-CW)	115,048 [114,975-115,122]	2023	48,745	82.0%
98	Solomon Islands Pijin (pis-Latn-SB)	101,588	2019	48,605	82.0%
99	Náhuatl de la Sierra noreste de Puebla (nah-sierra-noreste-puebla-Latn-MX)	134,737	2000	47,096	69.1%
100	Náhuatl de la Huasteca potosina (nah-huasteca-potosina-Latn-MX)	133,234	2000	46,643	65.2%

A.2 Standardized new-compute Top 100

This is a burden-per-compute planning order for one standardized from-scratch package. FECU means fresh-equivalent compute unit; it is not a monetary price or an observed complete-job token total. Because a common causal package effect is not measured, the model-median column divides the same common-prior unmet-need equivalent by modeled FECU rather than claiming observed access gained.

Efficiency rank	Need rank	Exact target	Unmet-need equivalent per million FECU	Standard compute (FECU)
1	1	Hindi (hi-Deva-IN)	97,776,202	1,220,000
2	2	Bahasa Indonesia (id-Latn-ID)	87,463,029	1,000,000
3	3	Russian - Russian Federation (ru-Cyrl-RU)	44,470,477	1,080,000
4	4	Punjabi (Pakistan standard) (pnb-Arab-PK)	33,919,207	1,140,000
5	5	Bengali (India standard) (bn-Beng-IN)	28,094,704	1,250,000
6	6	Marathi (mr-Deva-IN)	25,392,114	1,220,000
7	9	Javanese (jv-Latn-ID)	24,717,315	1,000,000
8	7	Telugu (te-Telu-IN)	24,265,996	1,250,000
9	8	Tamil (India standard) (ta-Taml-IN)	20,353,394	1,240,000
10	10	Central Thai (th-Thai-TH)	18,520,469	1,180,000
11	12	Pashto (Pakistan curriculum pack) (ps-Arab-PK)	16,985,284	1,140,000
12	16	Uzbek (uz-Latn-UZ)	16,913,981	1,000,000
13	11	Gujarati (gu-Gujr-IN)	16,607,535	1,220,000
14	14	Urdu (ur-Arab-IN)	16,444,539	1,140,000
15	13	Bhojpuri (bho-Deva-IN)	15,522,752	1,220,000
16	20	Spanish (es-Latn-US)	13,908,336	1,000,000
17	15	Burmese (my-Mymr-MM)	13,848,168	1,240,000
18	18	Sindhi (Pakistan standard) (sd-Arab-PK)	13,527,750	1,140,000
19	17	Kannada (kn-Knda-IN)	13,058,532	1,230,000
20	19	Moroccan Arabic (ary-Arab-MA)	13,051,502	1,140,000
21	23	Oromo (om-Latn-ET)	12,790,119	1,000,000
22	25	Standard Yoruba (yo-Latn-NG)	12,548,589	1,000,000
23	26	Sundanese (su-Latn-ID)	11,691,959	1,000,000
24	27	Standard (mg-Latn-MG)	11,558,560	1,000,000

Efficiency rank	Need rank	Exact target	Unmet-need equivalent per million FECU	Standard compute (FECU)
25	22	Saraiki (skr-Arab-PK)	11,271,279	1,140,000
26	21	Malayalam (ml-Mlym-IN)	10,513,470	1,240,000
27	24	Odia (or-Orya-IN)	10,351,703	1,230,000
28	30	Igbo (ig-Latn-NG)	10,300,842	1,000,000
29	28	Eastern Punjabi (pa-Guru-IN)	9,204,119	1,230,000
30	29	Amharic (am-Ethi-ET)	9,026,788	1,200,000
31	31	Urdu (ur-Arab-PK)	8,667,264	1,140,000
32	34	Kurmanji Kurdish (kmr-Latn-001)	7,458,209	1,000,000
33	32	Taiwan Mandarin (Guoyu) (cmn-Hant-TW)	6,282,643	1,450,000
34	37	isiZulu (zu-Latn-ZA)	6,036,704	1,000,000
35	33	Taiwanese Taigi - Southern Min (nan-Hant-TW)	5,551,623	1,450,000
36	35	Khmer (km-Khmr-KH)	5,223,619	1,230,000
37	40	Kazakh (kk-Cyrl-KZ)	5,134,148	1,080,000
38	36	Chhattisgarhi (hne-Deva-IN)	5,040,380	1,220,000
39	38	Nepali (Nepal standard) (ne-Deva-NP)	4,597,721	1,220,000
40	39	Assamese (as-Beng-IN)	4,437,836	1,250,000
41	43	Catalan (ca-Latn-ES)	4,307,845	1,000,000
42	41	Maithili (India) (mai-Deva-IN)	4,065,269	1,220,000
43	44	isiXhosa (xh-Latn-ZA)	3,980,641	1,000,000
44	42	Magahi (mag-Deva-IN)	3,859,217	1,220,000
45	45	Standard Kinyarwanda (rw-Latn-RW)	3,073,509	1,000,000
46	47	Madurese (mad-Latn-ID)	2,820,241	1,000,000
47	49	Turkmen (tk-Latn-TM)	2,657,005	1,000,000
48	50	Standard Somali (so-Latn-ET)	2,569,049	1,000,000

Efficiency rank	Need rank	Exact target	Unmet-need equivalent per million FECU	Standard compute (FECU)
49	46	Marwari (mwr-Deva-IN)	2,440,441	1,220,000
50	51	Sorani Kurdish (ckb-Arab-001)	2,208,197	1,140,000
51	52	Kyrgyz (ky-Cyrl-KG)	2,123,115	1,080,000
52	48	Cantonese (Yue) (yue-Hant-HK)	1,895,846	1,450,000
53	53	Tigrinya (ti-Ethi-ET)	1,817,361	1,200,000
54	54	Paraguayan Guarani (gug-Latn-PY)	1,629,362	1,000,000
55	56	Minangkabau (min-Latn-ID)	1,537,983	1,000,000
56	55	Georgian (ka-Geor-GE)	1,429,784	1,120,000
57	60	Banjarese (bjn-Latn-ID)	1,308,737	1,000,000
58	61	Buginese (bug-Latn-ID)	1,266,450	1,000,000
59	59	Armenian (hy-Armn-AM)	1,200,525	1,130,000
60	57	Awadhi (India) (awa-Deva-IN)	1,179,323	1,220,000
61	58	Maithili (Nepal) (mai-Deva-NP)	1,145,596	1,220,000
62	62	Brahvi (brh-Arab-PK)	1,051,572	1,140,000
63	63	Nepali (India jurisdiction pack) (ne-Deva-IN)	907,142	1,220,000
64	64	Bhojpuri (Nepal) (bho-Deva-NP)	632,875	1,220,000
65	65	Guinea-Bissau Kriol (pov-Latn-GW)	561,062	1,000,000
66	66	Q’eqchi’ (kek-GT)	520,136	1,000,000
67	68	K’iche’ (quc-GT)	499,097	1,000,000
68	69	Kreol Morisien (mfe-Latn-MU)	425,209	1,000,000
69	70	Basque (eu-Latn-ES)	407,030	1,000,000
70	67	Taiwan Hakka (hak-Hant-TW)	357,963	1,450,000
71	71	Awadhi (Nepal) (awa-Deva-NP)	304,770	1,220,000
72	72	Mam (mam-GT)	280,069	1,000,000

Efficiency rank	Need rank	Exact target	Unmet-need equivalent per million FECU	Standard compute (FECU)
73	73	Maayat'aan (yua-MX)	269,577	1,000,000
74	74	Shona (sn-Latn-ZA)	264,793	1,000,000
75	75	Standard Moroccan Amazigh (zgh-Tfng-MA)	202,960	1,250,000
76	76	Tseltal (tzh-MX)	201,861	1,000,000
77	77	Bolivian Aymara (ayr-BO)	197,684	1,000,000
78	78	Kaqchikel (cak-GT)	196,634	1,000,000
79	79	Welsh (cy-Latn-GB)	195,519	1,000,000
80	80	Aymara in Peru (ayr-PE)	193,950	1,000,000
81	81	Tsotsil (tzo-MX)	190,213	1,000,000
82	82	Russian mother-tongue population in Estonia (ru-Cyrl-EE)	163,429	1,080,000
83	83	Rwanda Kiswahili (swh-Latn-RW)	159,651	1,000,000
84	84	Tetun Prasa (tet-Latn-TL)	157,971	1,000,000
85	86	Maori (mi-Latn-NZ)	93,581	1,000,000
86	87	Belize Kriol (bzj-BZ)	86,924	1,000,000
87	88	Chol (ctu-MX)	85,943	1,000,000
88	89	Mambai (mgm-Latn-TL)	81,855	1,000,000
89	85	Magahi (mag-Deva-NP)	80,902	1,220,000
90	90	Q'anjob'al (kjb-GT)	70,940	1,000,000
91	91	Mexicano de la Huasteca hidalguense (nah-huasteca-hidalguense-Latn-MX)	69,358	1,000,000
92	92	Poqomchi' (poh-GT)	56,620	1,000,000
93	93	Makasae (mkz-Latn-TL)	53,586	1,000,000
94	94	Achi (acr-GT)	52,961	1,000,000
95	95	Diné Bizaad (nav-US)	51,686	1,000,000

Efficiency rank	Need rank	Exact target	Unmet-need equivalent per million FECU	Standard compute (FECU)
96	96	Ixil (ixl-GT)	50,776	1,000,000
97	97	Curaçao Papiamentu (pap-CW)	48,745	1,000,000
98	98	Solomon Islands Pijin (pis-Latn-SB)	48,605	1,000,000
99	99	Náhuatl de la Sierra noreste de Puebla (nah-sierra-noreste-puebla-Latn-MX)	47,096	1,000,000
100	100	Náhuatl de la Huasteca potosina (nah-huasteca-potosina-Latn-MX)	46,643	1,000,000

Appendix B. Stage, large-context, and explicit evidence-gap lanes

B.1 Stage opportunities

Stage rank	Exact opportunity	Cohort	Model median	Definition
1	Standard Written Chinese (zh-Hans-CN)	280,404,300	91,804,506	All formal-education enrolments reported by MOE in 2025; a stage-opportunity universe, not a count needing new Chinese material.
2	Standard Japanese (ja-Jpan-JP)	2,922,969	1,291,650	2,645,837 undergraduates plus 277,132 graduate/professional-degree students.

B.2 Mandatory large contexts and sentinel gaps

Display order	Population/context	Source-bounded measure	Evidence status	Required action
1	United States English-only home-language population age 5+	245,472,067 [245,250,680-245,693,454]	high for the published survey estimate and MOE; low for educational mastery or resource sufficiency	Acquire an exact comfortable-reader or stage-subject denominator and audit the functional supply endpoint before comparison.
2	English reported in first, second or third language position - India	128,539,090	high for the exact C-17 aggregation; low for current scale and academic proficiency	Acquire an exact comfortable-reader or stage-subject denominator and audit the functional supply endpoint before comparison.
3	Bengali reported in first, second or third language position - India	107,472,243	high for the exact C-17 aggregation; low for current scale, script literacy and academic proficiency	Acquire an exact comfortable-reader or stage-subject denominator and audit the functional supply endpoint before comparison.
4	Marathi reported in first, second or third language position - India	99,058,786	high for the exact 2011 C-17 aggregation; low for current threshold status and academic proficiency	Acquire an exact comfortable-reader or stage-subject denominator and audit the functional supply endpoint before comparison.

Display order	Population/context	Source-bounded measure	Evidence status	Required action
5	Telugu reported in first, second or third language position - India	94,501,603	high for the exact 2011 C-17 aggregation; low for current threshold status and academic proficiency	Acquire an exact comfortable-reader or stage-subject denominator and audit the functional supply endpoint before comparison.
6	English co-medium education exposure ceiling - Philippines	112,729,484	high for census population and legal medium; low for actual exposure, proficiency and resource sufficiency	Acquire an exact comfortable-reader or stage-subject denominator and audit the functional supply endpoint before comparison.
7	English education-language exposure ceiling - Pakistan	240,458,089	high for census universe and official curriculum role; low for actual exposure, proficiency and resource sufficiency	Acquire an exact comfortable-reader or stage-subject denominator and audit the functional supply endpoint before comparison.
8	English national education-language exposure ceiling - Nigeria	237,527,782	high for policy identity; medium for modeled population ceiling; low for actual English proficiency and resource sufficiency	Acquire an exact comfortable-reader or stage-subject denominator and audit the functional supply endpoint before comparison.
9	English secondary and higher-education exposure ceiling - Ethiopia	135,472,051	high for policy wording; medium for modeled population ceiling; low for actual stage exposure and proficiency	Acquire an exact comfortable-reader or stage-subject denominator and audit the functional supply endpoint before comparison.
10	French education-language exposure ceiling - Democratic Republic of the Congo	112,832,473	high for statutory language identity; medium for modeled population ceiling; low for actual proficiency and resource sufficiency	Acquire an exact comfortable-reader or stage-subject denominator and audit the functional supply endpoint before comparison.
1	French - DRC exposure context	not yet source-bounded	PRESENT_CONTEXT_STRATUM_NO_EXACT_EDITION_TARGET	Keep as context until tested or directly reported proficiency evidence exists
2	Algerian Amazigh varieties	not yet source-bounded	MISSING_EXACT_TARGET	Create exact variety and script targets before population binding
3	French - West and Central African education contexts	not yet source-bounded	PRESENT_BASE_LANGUAGE_ONLY_EXACT_TERRITORY_SCRIPT_OR_STANDARD_PROFILE_MISSING	Keep country-specific exposure and local-language alternatives visible
4	English - United States home-language stratum	not yet source-bounded	PRESENT_CONTEXT_STRATUM_NO_EXACT_EDITION_TARGET	Use as context and stage-specific negative control
5	English - India reported L1 L2 L3	not yet source-bounded	PRESENT_CONTEXT_STRATUM_NO_EXACT_EDITION_TARGET	Keep outside direct comfortable-reader ranking

Display order	Population/context	Source-bounded measure	Evidence status	Required action
6	English - Nigeria exposure ceiling	not yet source-bounded	PRESENT_CONTEXT_STRATUM_NO_EXACT_EDITION_TARGET	Do not treat national population as English users
7	English - Pakistan exposure ceiling	not yet source-bounded	PRESENT_CONTEXT_STRATUM_NO_EXACT_EDITION_TARGET	Do not treat national population as English users
8	English - Philippines exposure ceiling	not yet source-bounded	PRESENT_CONTEXT_STRATUM_NO_EXACT_EDITION_TARGET	Do not treat national population as English users
9	Hmong varieties in China	not yet source-bounded	MISSING_EXACT_TARGET	Create exact variety and script records before any person allocation

Appendix C. Structural access, signed-language, and displacement portfolios

C.1 Named sign-language and accessibility interventions

Display order	Target or intervention	Mode/tag	Territory/community	Denominator	Evidence status
1	Concise and extended text descriptions for meaningful instructional images	ACCESSIBILITY_AND_DELIVERY	Global; instantiate by language, curriculum, image type, and delivery format	not yet source-bounded	high_measure_medium_intervention_specificity
2	Downloadable born-accessible reflowable EPUB 3.3	ACCESSIBILITY_AND_DELIVERY	Global; instantiate and test by language and reading-system ecosystem	not yet source-bounded	high_standard_low_population_specificity
3	Downloadable full transcript for instructional audio or video	ACCESSIBILITY_AND_DELIVERY	Global; instantiate by media language, learner language, and subject	not yet source-bounded	high_measure_medium_intervention_specificity
4	Instructional video in one named national or regional sign language, paired with text and captions	ACCESSIBILITY_AND_DELIVERY	Named Deaf community and territory must be specified before implementation	not yet source-bounded	high_language_distinction_low_population_specificity
5	Localized Easy Read version with supportive images and reduced memory load	ACCESSIBILITY_AND_DELIVERY	Global; instantiate by language, cognitive-access context, subject, and community	not yet source-bounded	medium_low_due_to_dated_heterogeneous_meta_analysis
6	Localized audio description for instructional video and essential visuals	ACCESSIBILITY_AND_DELIVERY	Global; instantiate by media language, learner language, subject, and playback system	not yet source-bounded	high_measure_medium_intervention_specificity
7	Localized closed captions for prerecorded instructional media	ACCESSIBILITY_AND_DELIVERY	Global; instantiate by media language, learner language, and subject	not yet source-bounded	high_measure_medium_intervention_specificity
8	Localized live captions for synchronous instruction	ACCESSIBILITY_AND_DELIVERY	Global; instantiate by session language, learner language, and subject	not yet source-bounded	high_measure_medium_intervention_specificity
9	Localized plain-language version alongside the precise source	ACCESSIBILITY_AND_DELIVERY	Global; instantiate by language, literacy context, subject, and community	not yet source-bounded	medium_due_to_coarse_self_or_proxy_literacy_measure

Display order	Target or intervention	Mode/tag	Territory/community	Denominator	Evidence status
10	Native MathML for mathematical expressions	ACCESSIBILITY_AND_DELIVERY	Global; instantiate by language, locale, subject, browser, reading system, and assistive-technology stack	not yet source-bounded	high_standard_medium_interoperability_uncertainty
11	Navigable eBraille and embossable braille in a declared language and specialist code	ACCESSIBILITY_AND_DELIVERY	Global; instantiate by language, country convention, subject code, and available hardware	not yet source-bounded	high_code_evidence_low_population_specificity
12	Navigable full text synchronized with narration	ACCESSIBILITY_AND_DELIVERY	Global; instantiate by written/spoken language pair and learner community	not yet source-bounded	medium_due_to_high_meta_analysis_heterogeneity
13	Optional downloadable low-bitrate media variant preserving accessibility content	ACCESSIBILITY_AND_DELIVERY	Global; especially rural and low-income settings with intermittent or expensive connectivity	not yet source-bounded	high_constraint_evidence_low_population_specificity
14	Reflowed large-print output with user-controlled size, spacing, and contrast	ACCESSIBILITY_AND_DELIVERY	Global; instantiate by language, print standard, device, and low-vision user setting	not yet source-bounded	high_measure_medium_intervention_specificity
15	Self-contained downloadable offline accessible learning package	ACCESSIBILITY_AND_DELIVERY	Global; especially rural, low-income, displaced, home-learning, community-education, and field-based settings	not yet source-bounded	high_current_constraint_measure_medium_local_transferability
16	Semantic responsive HTML canonical content layer	ACCESSIBILITY_AND_DELIVERY	Global; instantiate and test by language, locale, device, and assistive-technology stack	not yet source-bounded	high_standard_low_population_specificity
17	Tactile graphics for spatial instructional diagrams	ACCESSIBILITY_AND_DELIVERY	Global; instantiate by curriculum, diagram type, braille convention, and production method	not yet source-bounded	high_functional_rationale_low_population_specificity
18	Brazilian Sign Language (Libras)	bzs-BR	Brazil	153,000	DIRECT_NAMED_SIGN_LANGUAGE_PERSON_MEASURE
19	British Sign Language (BSL)	bfi-GB	Scotland	117,300	DIRECT_NAMED_SIGN_LANGUAGE_PERSON_MEASURE
20	American Sign Language (ASL)	ase-CA	Canada	37,620	DIRECT_NAMED_SIGN_LANGUAGE_PERSON_MEASURE
21	New Zealand Sign Language (NZSL)	nzs-NZ	New Zealand	24,678	DIRECT_NAMED_SIGN_LANGUAGE_PERSON_MEASURE

Display order	Target or intervention	Mode/tag	Territory/community	Denominator	Evidence status
22	British Sign Language (BSL)	bfi-GB	England	20,739	DIRECT_NAMED_SIGN_LANGUAGE_PERSON_MEASURE
23	Auslan	asf-AU	Australia	16,242	DIRECT_NAMED_SIGN_LANGUAGE_PERSON_MEASURE
24	Irish Sign Language (IrSL)	isg-IE	Ireland	2,590	DIRECT_NAMED_SIGN_LANGUAGE_PERSON_MEASURE
25	British Sign Language (BSL)	bfi-GB	Wales	893	DIRECT_NAMED_SIGN_LANGUAGE_PERSON_MEASURE
26	British Sign Language (BSL)	bfi-GB	Northern Ireland	539	DIRECT_NAMED_SIGN_LANGUAGE_PERSON_MEASURE
27	American Sign Language (ASL)	ase-US	United States	not yet source-bounded	MEASUREMENT_GAP_NO_PERSON_DENOMINATOR
28	Chinese Sign Language (CSL)	csl-CN	China	not yet source-bounded	PROXY_NOT_A_SIGN_LANGUAGE_USER_COUNT
29	French Sign Language (LSF)	fsl-FR	Metropolitan France	not yet source-bounded	QUALIFIED_OR_UNDIFFERENTIATED_SIGN_MEASURE
30	German Sign Language (DGS)	gsg-DE	Germany	not yet source-bounded	PROXY_NOT_A_SIGN_LANGUAGE_USER_COUNT
31	Hawaiʻi Sign Language	hps-US-HI	Hawaiʻi, United States	not yet source-bounded	MEASUREMENT_GAP_NO_PERSON_DENOMINATOR
32	Indian Sign Language (ISL-India)	ins-IN	India	not yet source-bounded	PROXY_NOT_A_SIGN_LANGUAGE_USER_COUNT
33	Japanese Sign Language (JSL)	jsl-JP	Japan	not yet source-bounded	PROXY_NOT_A_SIGN_LANGUAGE_USER_COUNT
34	Quebec Sign Language (LSQ)	fcs-CA	Canada	not yet source-bounded	DIRECT_NAMED_SIGN_LANGUAGE_PERSON_MEASURE
35	Spanish sign language(s), undifferentiated	und-sign-ES	Spain	not yet source-bounded	QUALIFIED_OR_UNDIFFERENTIATED_SIGN_MEASURE

C.2 Host-specific displacement portfolios

Display order	Portfolio	Status cohort	Language/script arms	Evidence status
1	Sudan internal-displacement multi-language overlay	9,100,000	Sudanese Arabic; Dinka; Nuer; named Darfur and Nuba languages; each host-country language	DIRECT_STATUS_COHORT_DENOMINATOR_LANGUAGE_DISTRIBUTION_UNRESOLVED
2	Palestinian Arabic refugee-status overlay	6,000,000	Palestinian Arabic; Modern Standard Arabic; each host-country language	DIRECT_STATUS_COHORT_DENOMINATOR_LANGUAGE_DISTRIBUTION_UNRESOLVED
3	Syrian/North Levantine Arabic refugee overlay	4,900,000	Syrian/North Levantine Arabic; Modern Standard Arabic; each host-country language	DIRECT_STATUS_COHORT_DENOMINATOR_LANGUAGE_DISTRIBUTION_UNRESOLVED
4	Afghan displaced multi-language overlay	3,700,000	Dari; Pashto; Hazaragi; Uzbek and other explicitly measured Afghan languages; each host-country language	DIRECT_STATUS_COHORT_DENOMINATOR_LANGUAGE_DISTRIBUTION_UNRESOLVED
5	Rohingya stateless cohort - script interfaces	1,800,000	Rohingya Hanifi; Rohingya Latin; Rohingya Arabic; Bangla and Myanmar host/transition interfaces	DIRECT_STATUS_COHORT_DENOMINATOR_LANGUAGE_DISTRIBUTION_UNRESOLVED
6	Ukrainian-language/curriculum continuity cohort among displaced learners from Ukraine (language not enumerated)	700,000	Ukrainian Cyrillic continuity; Russian where directly measured; each host-country language	DIRECT_ENROLLED_LEARNER_COHORT_LANGUAGE_DISTRIBUTION_UNRESOLVED
7	DRC refugees and internal displacement - host-specific education portfolio	not yet source-bounded	French; Lingala; Kiswahili; Tshiluba; Kikongo varieties and other named languages; each host-country language	SOURCE_DISCOVERY_REQUIRED_NO_POPULATION_ASSIGNED
8	Somali refugees - host-specific education portfolio	not yet source-bounded	Somali Latin; Arabic where directly measured; each host-country language	SOURCE_DISCOVERY_REQUIRED_NO_POPULATION_ASSIGNED
9	South Sudanese refugees - host-specific education portfolio	not yet source-bounded	Juba Arabic; Dinka; Nuer; Bari and other named languages; each host-country language	SOURCE_DISCOVERY_REQUIRED_NO_POPULATION_ASSIGNED
10	Venezuelan refugees and migrants - host-specific education portfolio	not yet source-bounded	Venezuelan Spanish; Brazilian Portuguese and each other host-country language; Indigenous-language arms where directly evidenced	SOURCE_DISCOVERY_REQUIRED_NO_POPULATION_ASSIGNED

Appendix D. Bridge and interlanguage research portfolio

No entry receives family-wide population credit. Evidence grades order the display only; they are not benefit scores. Mathematics-specific comprehension, training burden, script, prior exposure, genre, and overlap with existing natural-language editions must be measured separately from shared-source production reuse.

Display order	Bridge or design	Class/mode	Evidence grade	Current disposition
1	Czech-Slovak receptive pair	Czech and Slovak Latin orthographies	EVIDENCE_GRADE_B	Strong natural-pair overlap and shared-source case.
2	Dutch-Afrikaans written receptive pair	Dutch and Afrikaans Latin orthographies	EVIDENCE_GRADE_B	Evidence-backed natural-pair aid.
3	Hindi-Urdu / Hindustani parallel registers	Devanagari; Perso-Arabic Nastaliq; Romanization only as optional navigation	EVIDENCE_GRADE_B	Strong shared-source parallel standards; reject one-script/one-lexicon substitution.
4	Mainland Scandinavian receptive multilingualism	national Latin orthographies; Bokmål and Nynorsk remain distinct labels	EVIDENCE_GRADE_B	Established overlap/parallel-source layer; not a new interlanguage.
5	Modern Standard Arabic (MSA)	Arabic script; accessible audio/plain-language or home-variety scaffolds are separate modalities	EVIDENCE_GRADE_B	Established learned-standard target with stage-specific bounds, not an interlanguage multiplier.
6	Spanish-Portuguese and Italian-Spanish receptive pairs	national Latin orthographies	EVIDENCE_GRADE_B	Natural-pair overlap layer; not a constructed interlanguage.
7	Interslavic	ordinary Slavic Latin; Cyrillic; extended etymological orthography only as optional research display	EVIDENCE_GRADE_C	Pilot/supplement; unranked as a family-wide language target until community-specific functional bounds exist.
8	Folksprak / whole-family Inter-Germanic hypothesis	draft Latin orthography; Hebrew-script Yiddish and other community orthographies are not covered	EVIDENCE_GRADE_D	Do not treat as current educational bridge; test separate natural-pair profiles instead.
9	Pan-Turkic constructed language / Common Turkic Alphabet	proposed 34-letter Latin alphabet; existing national Latin, Cyrillic and Perso-Arabic scripts remain separate arms	EVIDENCE_GRADE_D	Research infrastructure only; not a scored bridge-language edition.
10	Romance Neolatino	project Latin orthography with diacritics	EVIDENCE_GRADE_D	Research control/pilot only.

Display order	Bridge or design	Class/mode	Evidence grade	Current disposition
11	Bosnian-Croatian-Montenegrin-Serbian (BCMS) parallel standards	Bosnian/Croatian/Montenegrin Latin; Serbian Latin and Cyrillic	EVIDENCE_GRADE_B-/C+	Deployable shared-source parallel standards.
12	Esperanto	Latin alphabet with Esperanto diacritics	EVIDENCE_GRADE_B-/C for learning; D for zero-study reach	Learned-community target/control only.
13	Indonesian-Standard Malay pluricentric bridge	Rumi/Latin national standards; Jawi as separate Malaysian/Bruneian/community arm	EVIDENCE_GRADE_B-	Strong shared-source parallel-standard case, not one interlanguage.
14	Interlingua	Latin script	EVIDENCE_GRADE_D for zero-study reach	Learned-community target or research control; unranked for zero-study bridge reach.
15	International Sign	video/visual-gestural modality; captions, national sign-language versions and written text are separate modalities	EVIDENCE_GRADE_B-/C for conference use; D for universal education	Additional contact-layer summary only; never substitute for named sign-language editions.
16	Iranian Persian-Afghan Dari-Tajik parallel registers	Perso-Arabic for Iranian Persian/Dari; Tajik Cyrillic; transliteration as infrastructure	EVIDENCE_GRADE_B-	Strong shared-source, dual-script parallel-register case.
17	Kiswahili	standard Latin; Ajami only for separately evidenced communities	EVIDENCE_GRADE_B-/C+	Established learned natural-language bridge with territory-specific bounds.
18	Romance intercomprehension pedagogy	national Latin orthographies; Romanian diacritics; accessibility variants separately scoped	EVIDENCE_GRADE_B-	Taught access layer attached to natural editions; population effect remains partially identified.
19	Russian in post-Soviet Central Asia	Cyrillic; national-language Latin/Cyrillic/Perso-Arabic scripts remain separate	EVIDENCE_GRADE_B-/C+	Existing overlap/control layer, not a constructed bridge recommendation.
20	Turkish-Azerbaijani close-pair aids	Turkish Latin; Azerbaijani Latin; Iranian Azerbaijani Perso-Arabic only as a separate unevidenced arm	EVIDENCE_GRADE_B-	Pair-specific parallel aid/pilot, not a pan-Turkic edition.

Appendix E. Stage and subject recommendations

The 63 rows below are conditional package recommendations, not additional population ranks. They identify the first educational stages and material classes to investigate or commission for leading, stage-specific, regionally important, displaced, and structurally important routes. A recommendation does not claim that every listed resource is absent; its boundary note states the principal overreach to avoid. The full machine-readable register retains second-wave packages, delivery profiles, evidence keys, and confidence.

Target or route	Priority stages	First package classes	Why it may have outsized utility	Boundary
Standard Mandarin; Simplified Chinese (zh-Hans-CN)	postgraduate -> research -> undergraduate -> vocational-adult -> upper-secondary	research methods; reproducibility; open science; accessible STEM reference	large research community; open licensing and accessibility rather than assumed foundational scarcity	Do not infer national foundational need from population; audit rural migrant disability and field strata separately
Japanese (ja-Jpan-JP)	research -> postgraduate -> undergraduate -> accessibility	research reading/writing; open science; reproducibility; accessible advanced STEM	English-language research burden despite strong school outcomes	Current evidence points first to research and accessibility; it neither asserts whole-language primary scarcity nor excludes evidence-specific primary cohorts
Korean (ko-Kore-KR)	research -> postgraduate -> undergraduate -> accessibility	research methods; open science; accessible advanced STEM	advanced multilingual and open-access value in a strong school system	Do not claim foundational scarcity without stratum evidence
Standard Indonesian (id-Latn-ID)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate -> research -> early-childhood	mastery literacy/numeracy; complete mathematics/science; health; climate/disaster; teacher education	53% learning poverty; archipelagic disaster/climate; national bridge across regional languages	National Indonesian need is not reduced by any local translation activity
Javanese; Latin core (jv-Latn-ID)	early-childhood -> primary -> lower-secondary -> vocational-adult	oral/literacy/numeracy foundation; Indonesian bridge; health; disaster	large home-language stratum transitioning to Indonesian schooling	Do not collapse recorded dialects or substitute Javanese script without a separate literacy track
Sundanese; Latin core (su-Latn-ID)	early-childhood -> primary -> lower-secondary -> vocational-adult	foundation; Indonesian bridge; primary concepts; health	home-language foundation and national-language transition	Sundanese script is a separately scoped literacy/access edition

Target or route	Priority stages	First package classes	Why it may have outsized utility	Boundary
Madurese (mad-Latn-ID)	early-childhood -> primary -> lower-secondary -> vocational-adult	foundation; Indonesian bridge; health; disaster	home-language transition and island livelihood needs	Tag materially different eastern/western varieties
Banjarese (bjn-Latn-ID)	early-childhood -> primary -> vocational-adult -> lower-secondary	foundation; Indonesian bridge; health; wetland safety	river/wetland livelihoods and national-language transition	Banjar Kuala and Banjar Hulu require tagging
Buginese; Latin core (bug-Latn-ID)	early-childhood -> primary -> vocational-adult -> lower-secondary	foundation; Indonesian bridge; health and safety	maritime/agricultural livelihoods and bilingual transition	Do not silently mix Latin and Lontara literacy
Bangla (bn-Beng-BD)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate -> research -> early-childhood	foundation/catch-up; mathematics/science; health; teacher education; climate/disaster	51% learning poverty; delta/cyclone exposure; agriculture and climate-migration pressures	Complete Bangla ladder; retain English search terminology without making English a prerequisite
Standard Hindi (hi-Deva-IN)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate -> research	foundation/catch-up; mathematics/science; health; teacher education	massive foundation programme and large higher-education population	Do not use broad Hindi census grouping as proof of standard-Hindi comprehension
Marathi (mr-Deva-IN)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate	foundation/catch-up; mathematics/science; health	large exact language with independent school and academic pathway	Independent target; Hindi is not a coverage proxy
Telugu (te-Telu-IN)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate	foundation/catch-up; mathematics/science; health	large exact language and major higher-education/workforce community	Independent target; do not infer English proficiency from IT-sector visibility
Tamil (ta-Taml-IN)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate -> research	foundation/catch-up; mathematics/science; health	large school and research-language community with international diaspora	India and Sri Lanka locale profiles should be distinguished
Gujarati (gu-Gujr-IN)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate	foundation/catch-up; mathematics/science; health	large exact language with strong enterprise relevance	Independent target; Hindi is not coverage
Kannada (kn-Knda-IN)	primary -> lower-secondary -> upper-secondary -> undergraduate -> vocational-adult -> postgraduate	foundation/catch-up; mathematics/science; health	large exact language and major technology/higher-education region	Do not infer English comfort from regional industry profile
Malayalam (ml-Mlym-IN)	upper-secondary -> undergraduate -> postgraduate -> research -> vocational-adult -> primary	accessible STEM and health; research methods; finance/digital safety	strong school outcomes shift marginal value toward advanced/open/accessibility while retaining exact gap audit	Do not assume high literacy eliminates advanced-language burden

Target or route	Priority stages	First package classes	Why it may have outsized utility	Boundary
Odia (or-Orya-IN)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate	foundation/catch-up; mathematics/science; health; climate/disaster	large exact language; coastal/cyclone and agricultural applications	Independent target; Hindi is not coverage
Urdu (ur-Arab-PK)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate -> research	literacy/numeracy recovery; mathematics/science; health; teacher education	77% learning poverty; national bridge and later academic language	Urdu is national but not the home language of most early learners
Punjabi; Shahmukhi (pnb-Arab-PK)	early-childhood -> primary -> lower-secondary -> vocational-adult	foundation; Urdu bridge; health	home-language mismatch in a very large provincial population	Do not merge with Gurmukhi Punjabi output
Sindhi (sd-Arab-PK)	early-childhood -> primary -> lower-secondary -> upper-secondary -> vocational-adult	foundation; Urdu/English bridge; mathematics/science; health	home-language instruction plus climate-vulnerable livelihoods	Use the accepted Sindhi Arabic orthography; script variants are separate
Pashto (ps-Arab-PK)	early-childhood -> primary -> lower-secondary -> vocational-adult -> upper-secondary	foundation; Urdu bridge; health; teacher education	home-language mismatch and rural/conflict-affected strata	Pakistan locale; Afghan Pashto requires a separate curriculum/context profile
Saraiki (skr-Arab-PK)	early-childhood -> primary -> lower-secondary -> vocational-adult	foundation; Urdu bridge; health	home-language mismatch and rural livelihood relevance	Independent target; not a generic Punjabi or Urdu edition
Filipino; formal national register (fil-Latn-PH)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate -> research -> early-childhood	foundation recovery; mathematics/science; health; teacher education	91% learning poverty and severe secondary proficiency gaps	Formal Filipino and vernacular Tagalog should be register-tagged
Cebuano (ceb-Latn-PH)	early-childhood -> primary -> lower-secondary -> vocational-adult	foundation; Filipino/English bridge; health	large home-language stratum in a high-learning-poverty system	Do not treat census Bisaya and Cebuano labels as automatically identical
Hiligaynon (hil-Latn-PH)	early-childhood -> primary -> lower-secondary -> vocational-adult	foundation; Filipino/English bridge; health	large home-language stratum and transition burden	Separate from Cebuano/Bisaya
Ilocano (ilo-Latn-PH)	early-childhood -> primary -> lower-secondary -> vocational-adult	foundation; Filipino/English bridge; health	large home-language stratum and transition burden	Independent target
Burmese (my-Mymr-MM)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> health-professional -> research	portable catch-up; mathematics/science; health; teacher education	conflict/displacement and disrupted educational continuity	National language does not cover all minority-language entrants
Shan (shn-Mymr-MM)	early-childhood -> primary -> lower-secondary -> vocational-adult	foundation; Burmese bridge; health	minority-language and conflict/displacement access	Verify Shan orthography and locale; population basis remains uncertain

Target or route	Priority stages	First package classes	Why it may have outsized utility	Boundary
S'gaw Karen (ksw-Mymr-MM)	early-childhood -> primary -> lower-secondary -> vocational-adult	foundation; Burmese/host-language bridge; health	displacement and educational interruption	Do not merge S'gaw with Pwo Karen languages
Rohingya; Hanifi core (rhg-Rohg)	early-childhood -> primary -> lower-secondary -> vocational-adult -> health	foundation; host-language bridge; health; safety	displacement; limited formal continuity; health-information barriers	Hanifi Latin and Arabic-derived access editions must be separately tagged
Vietnamese (vi-Latn-VN)	upper-secondary -> vocational-adult -> undergraduate -> postgraduate -> research -> accessibility	secondary STEM; first-year quantitative bridge; research methods	stronger foundation shifts value toward advanced/open access and professional capacity	Do not use national outcomes to erase ethnic-minority early-language needs
Standard Thai (th-Thai-TH)	lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate -> research	mathematics/science progression; health; digital/financial capability	moderate national foundation gap and tertiary English bridge	National Thai is not evidence of coverage for Isan Pattani Malay Northern Thai or Northern Khmer
Isan language; Thai-script contemporary literacy (tts-Thai-TH)	early-childhood -> primary -> lower-secondary -> vocational-adult	oral/foundation scaffold; Standard Thai bridge; health	home-language versus national instructional-language transition; rural livelihoods	Do not claim full Lao or Standard Thai equivalence
Pattani Malay; Jawi core (mfa-Arab-TH)	early-childhood -> primary -> lower-secondary -> vocational-adult	foundation; Thai bridge; health	language-of-instruction transition and regional access	Malay varieties and scripts require explicit locale tags
Khmer (km-Khmr-KH)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate	foundation recovery; mathematics/science; health; teacher education	weak foundational learning plus low STEM/research participation	National Khmer does not cover Indigenous minority languages
Lao (lo-Laoo-LA)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate	foundation recovery; mathematics/science; health; teacher education	foundation and rural-connectivity gaps	National Lao does not automatically cover Khmu Hmong or other minority entrants
Kiswahili (sw-Latn-TZ-KE)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> teacher-education -> undergraduate -> postgraduate	foundation/catch-up; mathematics/science; health; teacher education	cross-border bridge; learning poverty; English tertiary barrier; rural livelihoods	Country roles differ; do not award blanket East African comprehension
Hausa; Latin core (ha-Latn-NG-NE)	early-childhood -> primary -> lower-secondary -> vocational-adult -> teacher-education	foundation; national-language bridge; health	large cross-border adult/health/agriculture reach in high-learning-poverty region	Latin and Ajami editions require separate script QA
Yoruba (yo-Latn-NG)	early-childhood -> primary -> lower-secondary -> vocational-adult -> upper-secondary	foundation; English bridge; health	large home-language stratum in high-learning-poverty setting	Independent target; do not infer English coverage
Igbo (ig-Latn-NG)	early-childhood -> primary -> lower-secondary -> vocational-adult -> upper-secondary	foundation; English bridge; health	large home-language stratum in high-learning-poverty setting	Dialect/standard choice requires explicit locale metadata

Target or route	Priority stages	First package classes	Why it may have outsized utility	Boundary
Nigerian Pidgin (pcm-Latn-NG)	vocational-adult -> health -> financial-digital -> lower-secondary-support	health; digital/financial safety; adult functional literacy	wide oral reach for actionable public and adult material	Supplement not substitute for exact school/home languages
Amharic (am-Ethi-ET)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> teacher-education	foundation/catch-up; mathematics/science; health; teacher education	large national bridge amid low connectivity and incomplete learning measurement	National role does not cover Oromo Somali Tigrinya or other exact languages
Oromo (om-Latn-ET)	early-childhood -> primary -> lower-secondary -> vocational-adult -> teacher-education	foundation; Amharic/English bridge; health	large regional language; rural and pastoralist relevance	Country and dialect profiles require tagging
Somali (so-Latn-SO)	early-childhood -> primary -> lower-secondary -> vocational-adult -> health -> teacher-education	foundation/catch-up; health; teacher education	conflict/displacement and recurrent drought	Somalia Kenya Ethiopia and diaspora curriculum contexts differ
Tigrinya (ti-Ethi-ET-ER)	early-childhood -> primary -> lower-secondary -> vocational-adult -> health	foundation; national/host-language bridge; health	conflict/displacement and rural access	Ethiopia and Eritrea locale/curriculum profiles should be separate
Modern Standard Arabic (ar-Arab-MENA)	early-childhood-bridge -> primary -> lower-secondary -> upper-secondary -> vocational-adult -> undergraduate -> postgraduate -> research	language-rich MSA bridge; literacy/numeracy; mathematics/science; health	more than half of MENA children in learning poverty; MSA/home-variety distance	MSA is a shared formal register not proof that one edition fully serves every country or colloquial community
Moroccan Arabic; Darija (ary-Arab-MA)	early-childhood -> primary -> vocational-adult -> health -> lower-secondary-support	oral/foundation scaffold; MSA/French bridge; health	distance between home variety and formal school/academic languages	Supplement to MSA not automatic substitute; regional variety must be tagged
Iranian Persian (fa-Arab-IR)	upper-secondary -> undergraduate -> postgraduate -> research -> vocational-adult	accessible STEM; research methods; open science; health	large non-English research and tertiary population	Do not treat Persian as automatic Dari or Tajik coverage
Dari (prs-Arab-AF)	early-childhood -> primary -> lower-secondary -> vocational-adult -> health -> upper-secondary	portable foundation/catch-up; health; safety	conflict/displacement and academic-language interruption	Separate from Iranian Persian and Tajik; preserve locale terminology
Afghan Pashto (ps-Arab-AF)	early-childhood -> primary -> lower-secondary -> vocational-adult -> health -> upper-secondary	portable foundation/catch-up; health; safety	conflict/displacement and academic-language interruption	Separate curriculum/context profile from Pakistan Pashto
Kurmanji Kurdish; Latin script (ku-Latn-TR)	early-childhood -> primary-support -> vocational-adult -> health -> secondary-support	home-language literacy; Turkish/host-language bridge; health	minority/displacement and host-language barriers	Do not merge with Sorani or other Kurdish languages/scripts
Sorani Kurdish (ckb-Arab-IQ)	early-childhood -> primary -> lower-secondary -> vocational-adult -> undergraduate	foundation; Arabic/Persian bridge; health	distinct school/public language with regional academic-language bridges	Independent from Kurmanji; country locale matters

Target or route	Priority stages	First package classes	Why it may have outsized utility	Boundary
Haitian Creole; Kreyòl (ht-Latn-HT)	early-childhood -> primary -> lower-secondary -> vocational-adult -> health -> upper-secondary -> undergraduate	complete foundation/catch-up; French bridge; mathematics/science; health	home-language/national academic-language divide and disaster exposure	Kreyòl is not simplified French and French is not automatic coverage
Ayacucho Quechua (quy-Latn-PE)	early-childhood -> primary -> vocational-adult -> health -> teacher-education	community-language literacy/numeracy; Spanish bridge; health	Indigenous language and rural health/agriculture access	One exact Quechua variety; never label as all Quechua
Paraguayan Guaraní (gn-Latn-PY)	early-childhood -> primary -> lower-secondary -> vocational-adult -> health	foundation; Spanish bridge; health	large bilingual population with distinct home/public language roles	Paraguayan standard does not cover all Guaraní varieties
Indian Sign Language (ins-IN)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> health -> undergraduate	video-native mathematics/science; health; digital/financial safety	documented demand and existing secondary/senior-secondary precedent	Video-native sign language is not a word-for-word Hindi or English derivative
Chinese Sign Language (csl-CN)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> health -> undergraduate	video-native mathematics/science; health; digital/financial safety	education and employment access barriers associated with hearing loss	Regional CSL varieties and written-Chinese literacy profiles require tagging
Brazilian Sign Language; Libras (bzs-BR)	primary -> lower-secondary -> upper-secondary -> vocational-adult -> health -> undergraduate	video-native mathematics/science; health; digital/financial safety	education and employment access barriers associated with hearing loss	Libras is independent from Portuguese and other sign languages
Spanish; Latin America regional profile (es-Latn-419)	research -> postgraduate -> undergraduate -> vocational-adult -> accessibility	research methods; open science; accessible advanced STEM	large existing ecosystem shifts marginal value toward open advanced and specialist gaps	Negative control for generic foundation; country and Indigenous-language strata remain separate
French (fr-Latn-FR)	research -> postgraduate -> accessibility	open science; reproducibility; accessible advanced reference	resource-rich control; use only field-specific verified gaps	Do not use metropolitan French coverage as proof for African home-language access
German (de-Latn-DE)	research -> postgraduate -> accessibility	open science; reproducibility; accessible advanced reference	resource-rich control; use only field-specific verified gaps	Negative control for generic school translation
Russian (ru-Cyrl-RU)	research -> postgraduate -> undergraduate -> accessibility	open science; current methods; accessible advanced STEM	large scientific-language ecosystem with current/open/accessibility gaps to audit	Do not treat Russian as automatic coverage of Ukrainian Central Asian or minority languages

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